3. Cell organelles & functions

Mitochondria: Ultrastructure, aerobic metabolism, Oxidative Phosphorylation, electron transport and complexes

Chloroplast: Ultrastructure, chloroplast metabolism, photosynthesis electron transport

Ribosome: Ultrastructure, ribosome subunits, polysomes, protein synthesis, SRP-ribosome complex

Centrioles & basal bodies: Cilia & Flagella: Distribution, Structure of cilia & flagella, Chemical composition, Origin of cilia, Physiology of ciliary movement

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Evolution of aerobes

The earth is 4.6 billion years (BY) old. In ancient times (two billion years ago) the Earth consisted of reduced molecules like Hydrogen, ammonia, and water. So the earth of this period was populated by anaerobes (bacteria and archeobacteria)- organisms that captured and utilized energy through oxygen-independent metabolism, such as glycolysis and fermentation. Archeobacterai uses a modified form of glycolysis which involves the use of carbon dioxide as an electron acceptor to oxidize hydrogen to give methane. Methanogenesis uses a range of coenzymes that are unique to these archaea, such as coenzyme M and methanofuran.

Cyanobacteria appeared – Then, a new kind of organism that carried out a new type of photosynthetic process, one in which water molecules were split apart and molecular oxygen was released. The earth's environment began to accumulate significant levels of oxygen, which set the stage for a dramatic change in the types of organisms that would come to inhabit the planet.

The species which possessed metabolic pathways that utilized the oxygen evolved rapidly.

Without the ability to use oxygen, organisms could only extract a limited amount of energy from their foodstuffs, excreting energy-rich products such as lactic acid and ethanol, which were unable to metabolize further. In contrast, organisms that incorporated O2 into their metabolism could completely oxidize such compounds to CO2 and H2O and, in the process, extra a much larger percentage of their energy content. These organism that became dependent on oxygen were the earth's first aerobes and they gave rise to all the oxygen-dependent prokaryotes and eukaryotes living today.

Prokaryote Habitats

Prokaryotes have a wide range of metabolisms. So, they have wide range of habitat.

An **anaerobic organism** is any organism that does not need oxygen for growth and even dies in its presence. **Obligate anaerobes** will die when exposed to atmospheric levels of oxygen. *Clostridium perfringens* bacteria, which are commonly found in soil around the world, are obligate anaerobes. The main types of bacteria in the colon are obligate anaerobes. Obligate anaerobes use molecules other than oxygen as terminal electron acceptors. Bacteria in colon serve a central line of resistance to colonization by exogenous microbes and thus assist in preventing the potential invasion of the intestinal mucosa by an incoming pathogen. They help in break down of nondigestible carbohydrates include large polysaccharides (i.e., resistant starches, pectins, and cellulose). They also synthesize vit k and vit B.

Organisms that are **obligate aerobes** need oxygen to live. Obligate aerobes are found only in places with molecular oxygen. They use oxygen as a terminal electron acceptor while making ATP. *Mycobacterium tuberculosis* is an obligate aerobe bacteria requiring oxygen in its metabolism processes. Because of this oxygen requirement, *M. tuberculosis* manifests in the lung of mammals that have very high volume of oxygen. The optimum growth condition for this bacterium is at 37°C, pH 6.4–7.0, and oxygen level of >95%. *Mycobacterium tuberculosis* bacteria. causes tuberculosis (TB).

Prokaryote Habitats

Facultative anaerobic organisms, which are usually prokaryotic, make ATP by aerobic respiration if oxygen is present, but can also survive without oxygen. In the absence of oxygen, they switch to the process of fermentation to make ATP. **Fermentation** is a type of heterotrophic metabolism that uses organic carbon instead of oxygen as a terminal electron acceptor. Examples of facultative anaerobic bacteria are the *Staphylococci* species, *Escherichia coli, Corynebacterium* species, and *Listeria* species. Many bacteria that cause human diseases are facultatively anaerobic.

Many prokaryotes, small simple cells like bacteria, can perform aerobic cellular respiration using cell membrane. They **move electrons back and forth across their cell membrane.** Bacteria lack intracellular compartments, called organelles, that other cells have. Thus, in bacteria cellular respiration takes place in the cytoplasm and at the plasma membrane.

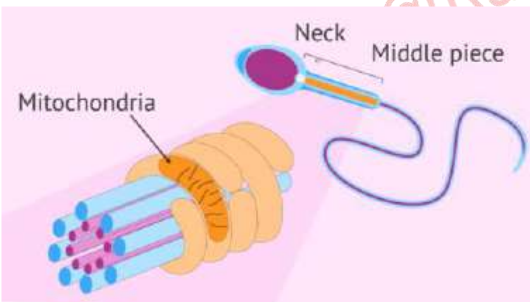
In eukaryotes the utilization of oxygen as a mean of energy **extraction** takes place in a specialized organelle, the mitochondria

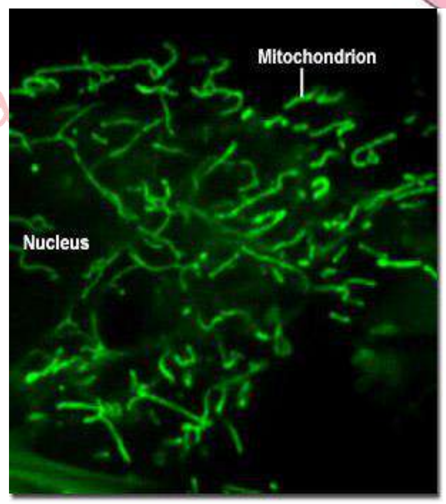
Assignment: PowerPoint presentation
Bacterial respiration (Roll no: 26-28)

Content:
Figure and process

Mitochondria: Structure

Mitochondria are visible under the light microscope although little detail can be seen. Detailed structure can be seen from electron microscope. Its structure varies depending upon the cell type. The hepatocyte mitochondria appear as bean-shaped (ovoids) organelles ranging form 1 to 4 µm in length. Fibroblast mitochondria are usually long filaments (1 to 10 µm in length). It can appear as a highly branched, interconnected tubular network in sperm cells. Their shapes change continually through the combined actions of fission, fusion, and motility.



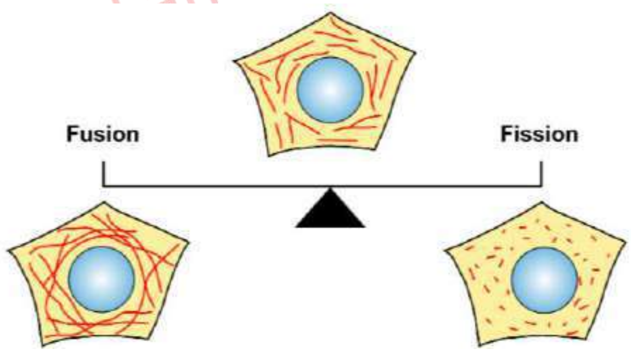


Fibrocytes

Fission and fusion of mitochondria:

Mitochondria are highly dynamic, complex organelles that continuously alter their shape, ranging between two opposite processes, fission and fusion, in response to several stimuli and the metabolic demands of the cell. To maintain the integrity and good working condition, mitochondrial fusion and fission are crucial. Therefore, deficits in either mitochondrial fusion or fission can also cause problems in different cells and the one of them is neuronal cell which may lead to different human neurological disorders. It is also necessary for the proper functioning of brain.

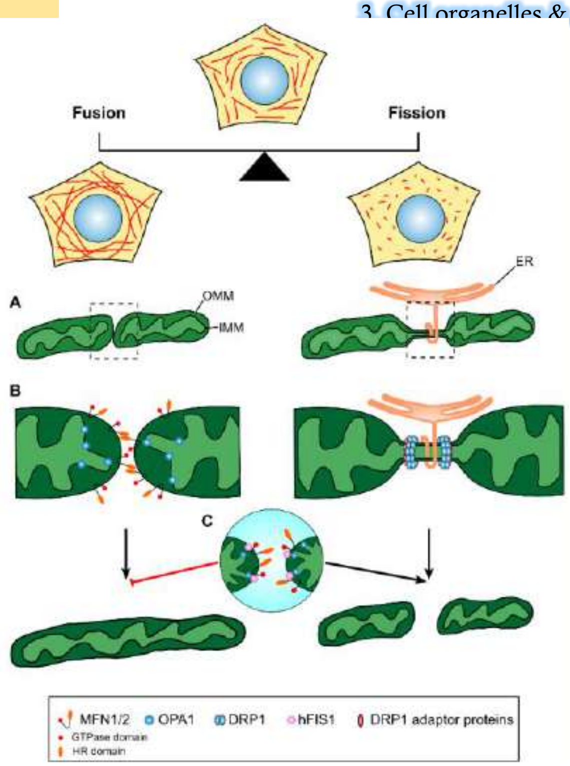
Mitochondria can fuse with one another, or split in two. When fusion becomes more frequent than fission, the mitochondria tend to become more elongated and interconnected. When fission is more predominance than fusion, it leads to the formation of more numerous and distinct mitochondria. The balance between fusion and fission is needed for healthy cell.



Fission and fusion of mitochondria:

Fusion between mitochondrial outer membranes is mediated by membrane-anchored proteins/receptors named mitofusin (Mfn1/2) and optic atrophy 1 (OPA1) in mammals. In a research, defect in such proteins has been observed in nonneurological disorders such as obesity, diabetes, vascular proliferative disorders, and cardiomyopathy. Although further study is required, the above study show some connection of defect of mitochondrial fusion and pathogenesis.

The **mitochondrial fission** is apparently induced by endoplasmic reticulum (ER). The ER tubule appear to initiate constriction , which then completed through the action of soluble proteins gathering on the outer surface of the mitochondria from the cytosol. The cytosolic protein Drp1 help to form spirals around mitochondria that constrict to sever both inner and outer membranes.



Function of Mitochondria:

1. ATP Generation:
   Mitochondria occupy 15 to 20 percent of the volume of an average mammalian liver cell and contain more than a thousand different proteins. It is estimated that in liver, mitochondria constitute 30 to 35 % of the total protein content of the cell and in kidney, 20 %. They are known for generating ATP. The major raw material for this energy is fatty acid-containing oil droplets which it will oxidize. In plant they are primary supplier of ATP in nonphotoxynthetic tissues, as well as being a source of ATP in photosynthetic leaf cells during periods of dark.

2. Mitochondria synthesizes certain amino acids, heme groups,
3. They play role in uptake and release of calcium ions (regulating the Ca+2 concentration of cytosol). Calcium signalling is very dynamic and must be tightly regulated; a release of calcium ions into the cytoplasm couples to many downstream effects in cells, such as cell division. Too much cytosolic calcium results in apoptosis, which is programmed cell death
4. In apoptosis of multicellular animal, it plays a vital role. The are distributed within the cytoplasm..
5. They can move freely, carrying ATP where needed, but in others they are located permanently near the region of the cell where presumably more energy in needed. A normal liver cell contain between 1000 and 1600 mitochondria. In cancer the number of mitochondria diminish due to decreased oxidation that accompanies the increase to anaerobic glycolysis in cancer. An increased number of mitochondria has also been found in human hyperthyroidism.

Plant cell have less mitochondria than animal cells. Why?

Mitochondria: Structure

Mitochondria consists of two membrane and two compartment..

Membranes

Both the inner and the outer mitochondrial membranes resemble the plasma membrane in molecular structure. They are composed of two layers of phospholipid molecule.

The section between the outer and inner membrane is called the inter-membrane space. It is about 6 to 8 nm wide. It contains a clear homogeneous fluid. The inner membrane projects into the mitochondrial cavity forming the infoldings called cirstae. The aqueous compartment formed by the inner membrane within the interior of the mitochondrion is called mitochondrial matrix. The matrix has a gel-like consistency owing to the presence of high concentration of water-soluble proteins.

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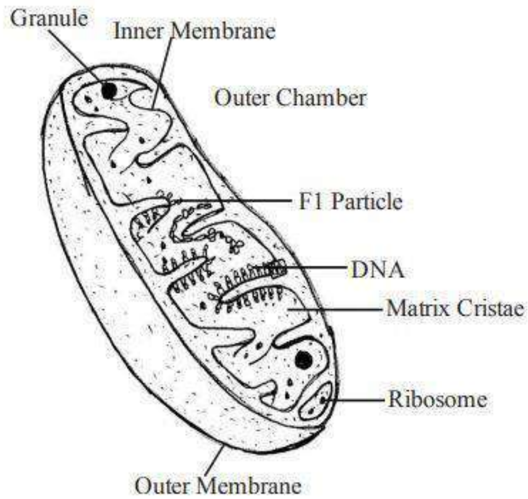


Fig: Structure of mitochondria organelle

Mitochondria: Structure

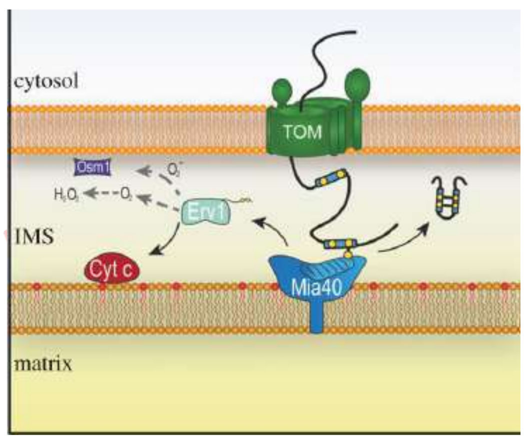
Mitochondria consists of two membrane and two compartment..

1. **Outer Membrane**

The outer membrane is smooth permeable to most small molecules, having trans-membrane channels formed by the protein 'porin'. It consists of about 50% lipid, including a large amount of cholesterol. It contains some enzymes but is poor in protein.

2. **Inner Membrane**

The inner membrane is selectively permeable and regulates the movement of materials into and out of the mitochondrion. It is rich in enzymes and carrier proteins permease. It contains ATP synthase which generates ATP in the matrix, and transport proteins that regulate the movement of metabolites into and out of the matrix. It has a very high protein/lipid ratio (about 4:1 by weight). It lacks cholesterol.



Cardiolipin is closely associated with certain integral proteins and is apparently required for their activity. Cardiolipin (CL) is a unique phospholipid which is localized and synthesized in the inner mitochondrial membrane (IMM). The inner membrane is arranged into cristae in order to increase the surface area available for energy production via oxidative phosphorylation.

Mitochondria: Structure

Matrix

The space between the cristae called the inner chamber is filled with a gel like material termed the mitochondrial matrix. It contains proteins, lipids, some ribosomes, RNA, one or two DNA molecules and certain fibrils, crystals and dense granules. The ribosome are smaller than that of found in cytosol. Thus it has its own genetic material and the machinery to manufacture their own RNAs and protein. The enzymes present in the matrix play an important role in the synthesis of ATP molecules.

Mitochondria replicate their DNA by a process called **binary fission** and can use this to make multiple copies in one mitochondrion.
Their DNA has **maternal lineage** which means their DNA is passed from mother to child with little change.
Mitochondrial DNA contains 37 genes, all of which are essential for normal mitochondrial function. Thirteen of these genes provide instructions for making enzymes involved in oxidative phosphorylation. The remaining genes provide instructions for making molecules called transfer RNA (tRNA) and ribosomal RNA (rRNA), which are required for protein synthesis from the amino acids.

The crista lumen contains large amounts of the small soluble electron carrier protein cytochrome *c*.

Health condition due to damage in mitochondrial DNA.

1. Age related Hearing-loss:

Changes in mitochondrial DNA are among the best-studied genetic factors associated with age-related hearing loss. It typically affects both ears and worsens gradually over time, making it difficult to understand speech and hear other sounds.
As people age, mitochondrial DNA suffers form different kinds of damages which result in deletions and other changes. This damages are specially due to the harmful molecules called reactive oxygen species (ROS), which are byproducts of energy production(electron transport chain) in mitochondria. Damage to proteins in the electron transport chain may worsen the situation by producing even more ROS. These reactive oxygen species (ROS) damage proteins, lipids, and DNA. Sometime such reactive oxygen species are produced in the body form other processes.). Mitochondria use quality-control proteases to eliminate damaged proteins. Mitochondrial DNA is especially vulnerable because it has a limited ability to repair itself. As a result, reactive oxygen species easily damage mitochondrial DNA, causing cells to malfunction and ultimately to die. Cells that have high energy demands, such as those in the inner ear that are critical for hearing, are particularly sensitive to the effects of mitochondrial DNA damage. This damage can irreversibly alter the function of the inner ear, leading to hearing loss.

2. Cytochrome c oxidase deficiency

It is initiated when there is damage in the mitochondrial genes associated for the production of cytochrome c oxidase. The major role of cytochrome c oxidase (a large enzyme complex) is in the last step in oxidative phosphorylation before the generation of ATP. The mtDNA mutations or damage may lead to change or alteration in the structure of cytochrome c oxidase. As a result, cytochrome c oxidase cannot function. A lack of functional cytochrome c oxidase disrupts oxidative phosphorylation, causing a decrease in ATP production. Researchers believe that impaired oxidative phosphorylation can lead to cell death in tissues that require large amounts of energy, such as the brain, muscles, liver and heart.

Fate of Glucose metabolism:

Glucose metabolism is a primary source of energy and biomaterials for the maintenance of life. Hexokinase (HK) phosphorylates glucose to glucose-6-phosphate (G6P). The product generally enters into three metabolic pathway:

a. Pentose phosphate pathway (PPP): PPP plays an important role in the synthesis of nucleic acids for DNA and RNA, as well as generation of NADPH for the synthesis of lipids and maintenance of intracellular redox homeostasis.
b. Glycogenesis: synthesis of glycogen

c. glycolytic pathway, generating NADH, ATP, and pyruvate .In the presence of sufficient oxygen aerobic metabolism occurs, in which the generated pyruvate is fed into mitochondria and fully oxidized to produce more ATP. When oxygen is limited, anaerobic metabolism occurs, which converts pyruvate to lactic acid (lactate). In anaerobic metabolism glycolysis becomes the main source for ATP production. However the numbers of ATP production is very less compared to aerobic metabolism.

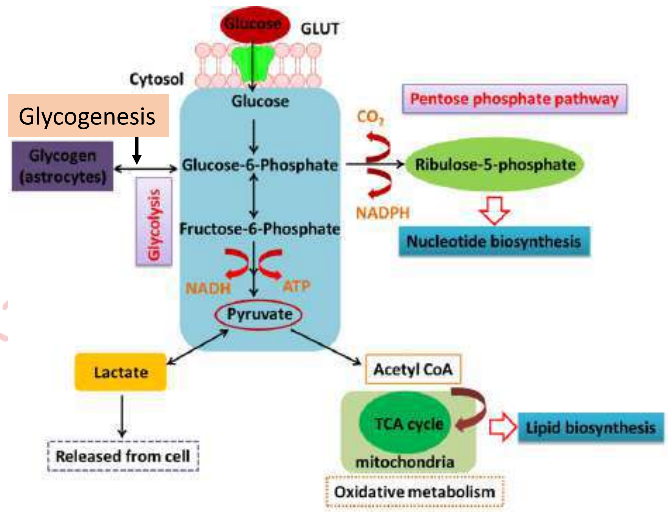


Fig: Fate of Glucose metabolism to maintain homeostasis

Glycolysis:

Glycolysis is the first step in the breakdown of glucose to extract energy for cellular metabolism. Nearly all living organisms carry out glycolysis as part of their metabolism. The process does not use oxygen and is therefore **anaerobic**. Glycolysis takes place in the cytoplasm of both prokaryotic and eukaryotic cells.
Glucose enters heterotrophic cells in two ways. One method is through secondary active transport in which the transport takes place against the glucose concentration gradient. The other mechanism uses a group of integral proteins called GLUT proteins, also known as glucose transporter proteins. These transporters assist in the facilitated diffusion of glucose.

Glycolysis consists of two distinct phases.

First Half of Glycolysis (Energy-Requiring Steps)
Step 1. The first step in glycolysis is catalyzed by hexokinase, an enzyme with broad specificity that catalyzes the phosphorylation of six-carbon sugars. Hexokinase phosphorylates glucose using ATP as the source of the phosphate, producing glucose-6-phosphate, a more reactive form of glucose. This reaction prevents the phosphorylated glucose molecule from continuing to interact with the GLUT proteins, and it can no longer leave the cell because the negatively charged phosphate will not allow it to cross the hydrophobic interior of the plasma membrane.

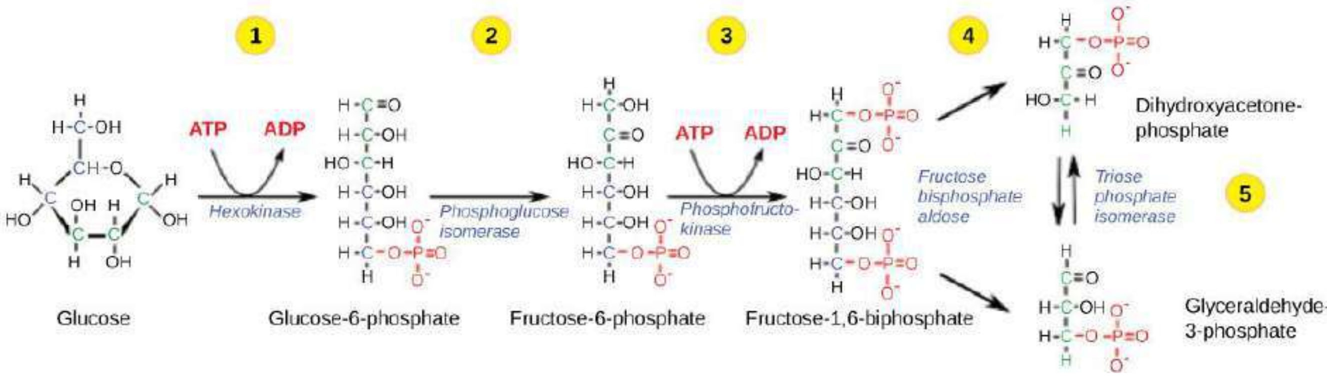
Step 2. An isomerase enzyme converts glucose-6-phosphate into one of its isomers, fructose-6-phosphate

Step 3. The third step is the phosphorylation (using ATP) of fructose-6-phosphate, catalyzed by the enzyme phosphofructokinase . A second ATP molecule donates a high-energy phosphate to fructose-6-phosphate, producing fructose-1,6-bisphosphate. In this pathway, phosphofructokinase is a rate-limiting enzyme. It is active when ATP is low and ADP is high and vice versa. Thus, if there is "sufficient" ATP in the system, the pathway slows down. This is a type of end product inhibition, since ATP is the end product of glucose catabolism.

Step 4. The fourth step in glycolysis employs an enzyme, aldolase, to cleave 1,6-bisphosphate into two three-carbon isomers: dihydroxyacetone-phosphate and glyceraldehyde-3-phosphate.

Step 5. In the fifth step, an isomerase transforms the dihydroxyacetone-phosphate into its isomer, glyceraldehyde-3-phosphate. Thus, the pathway will continue with two molecules of a single isomer.

The first half of glycolysis uses two ATP molecules in the phosphorylation of glucose,



Second Half of Glycolysis (Energy-Releasing Steps)

In the first half, glycolysis has cost the cell two ATP molecules generating two, three-carbon sugar molecules. In the second half both of these molecules will generate sufficient energy to pay back the two ATP molecules used as an initial investment and produce a profit for the cell of two additional ATP molecules and two even higher-energy NADH molecules.

Step 6. The sixth step in glycolysis oxidizes the glyceraldehyde-3-phosphate extracting high-energy electrons, which are picked up by the electron carrier NAD^+, producing NADH. The sugar is then phosphorylated by the addition of a second phosphate group, producing 1,3-bisphosphoglycerate. Note that the second phosphate group does not require another ATP molecule.

Step 7. In the seventh step, catalyzed by phosphoglycerate kinase (an enzyme named for the reverse reaction), 1,3-bisphosphoglycerate donates a high-energy phosphate to ADP, forming one molecule of ATP.

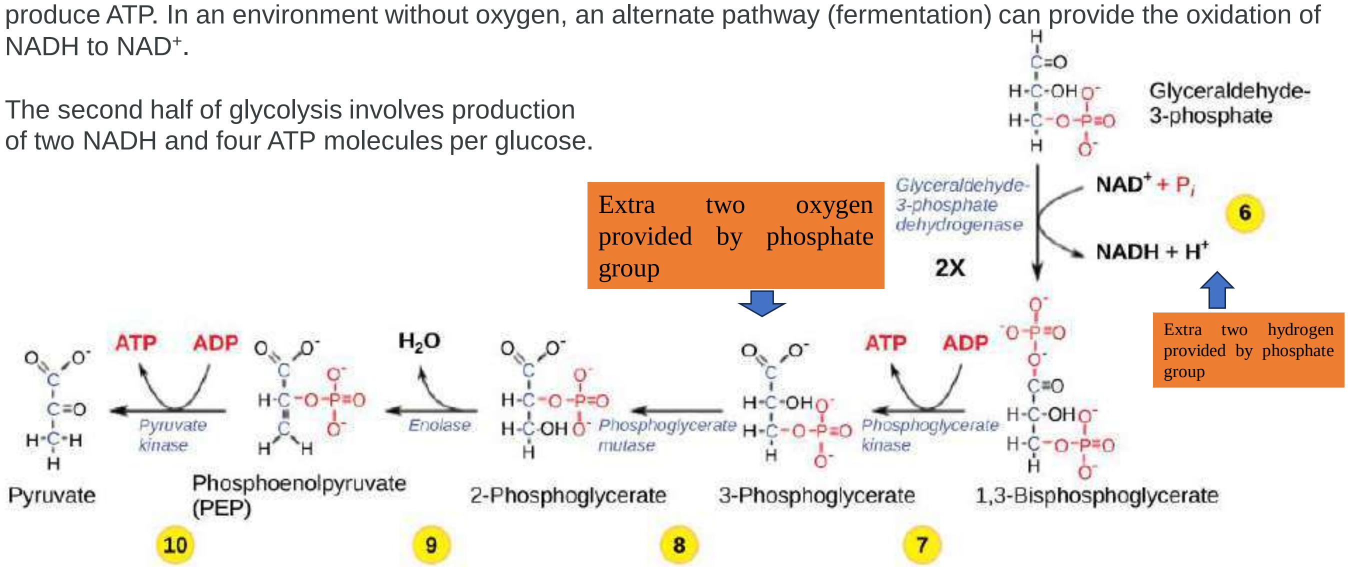
Step 8. In the eighth step, the remaining phosphate group in 3-phosphoglycerate moves from the third carbon to the second carbon, producing 2-phosphoglycerate (an isomer of 3-phosphoglycerate). The enzyme catalyzing this step is a mutase (isomerase).

Step 9. Enolase catalyzes the ninth step. This enzyme causes 2-phosphoglycerate to lose water from its structure; this is a dehydration reaction, resulting in the formation of a double bond that increases the potential energy in the remaining phosphate bond and produces phosphoenolpyruvate (PEP).

Step 10. The last step in glycolysis is catalyzed by the enzyme pyruvate kinase and results in the production of a second ATP molecule by substrate-level phosphorylation and the compound pyruvic acid (or its salt form, pyruvate).

The second half also contain potential limiting factor for this pathway. The continuation of the reaction depends upon the availability of the oxidized form of the electron carrier, NAD^+. Thus, NADH must be continuously oxidized back into NAD^+ in order to keep this step going. If NAD^+ is not available, the second half of glycolysis slows down or stops. If oxygen is available in the system, the NADH will be oxidized readily, in ETC reaction to produce ATP. In an environment without oxygen, an alternate pathway (fermentation) can provide the oxidation of NADH to NAD^+.

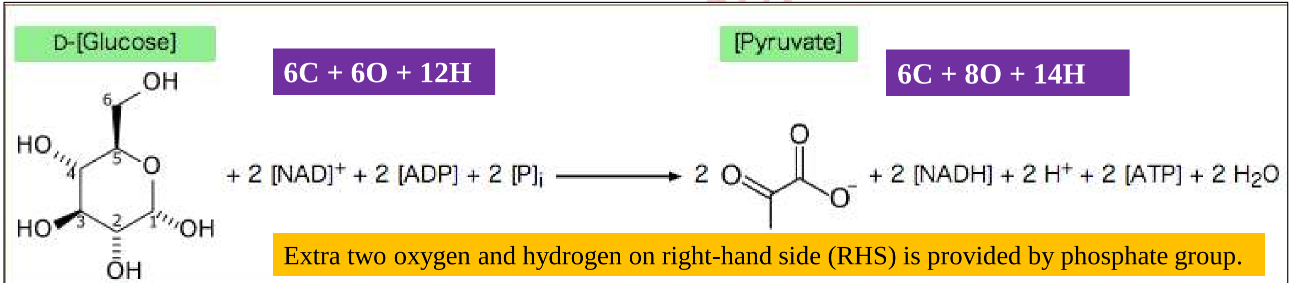
The second half of glycolysis involves production of two NADH and four ATP molecules per glucose.



Products of Glycolysis

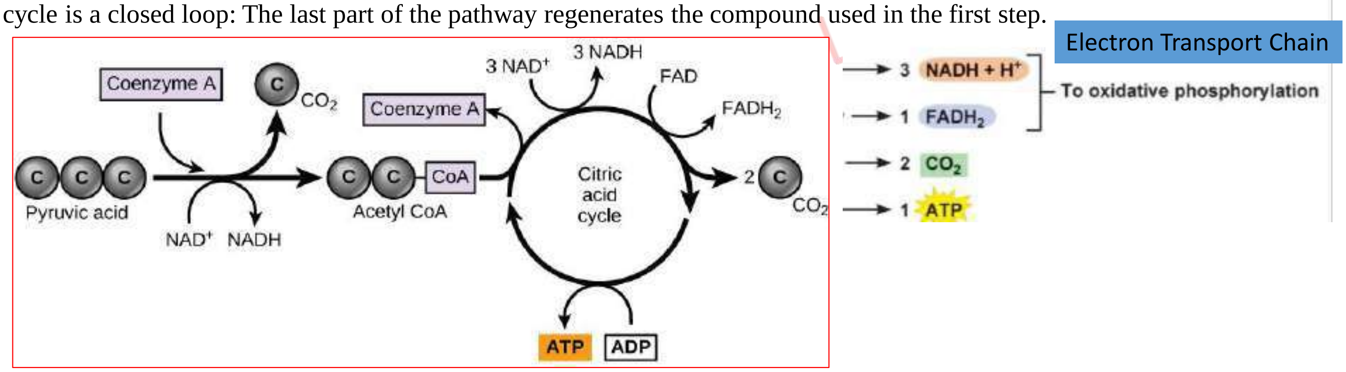
Glycolysis takes place in the Cytoplasm of the cell. Two ATP are consumed while four ATP and two NADH are produced. So, in glycolysis, one molecule of glucose is oxidized to two molecules of pyruvate with the process giving rise to a net gain of two ATP molecules , in addition, two molecules of NADH. Each NADH will give 3 molecules of ATP when further oxidized in mitochondria through electron transport chain. If the pyruvate turns to lactic acid due to lake of oxygen (called anaerobic metabolism) then no further ATP are generated. So, the final energy of anaerobic metabolism in mammal or in prokaryotes (fermentation) is equivalent to 2 ATP molecule.

$$\text{Glucose} + 2NAD^+ + 2P_i + 2ADP \rightarrow 2\text{Pyruvate} + 2NADH + 2ATP + 2H^+ + 2H_2O$$



TCA/Citric Acid/Krebs Cycle

In eukaryotic cells, the pyruvate molecules produced at the end of glycolysis are transported into mitochondria, which are sites of cellular respiration. If oxygen is available, aerobic respiration will go forward. In mitochondria, pyruvate will be transformed into a two-carbon acetyl group (by removing a molecule of carbon dioxide) that will be picked up by a carrier compound called coenzyme A (CoA), which is made from vitamin B5. The resulting compound is called acetyl CoA. The acetyl CoA then enter the TCA by attachment to oxaloacetate to form the six-carbon citrate molecule. In a stepwise series of oxidative reactions, the citrate is converted back to oxaloacetate (the cycle). The theoretical yield from each pyruvate is 2 moles of CO2, 3 moles of NADH, 1 mole of FADH2, and 1 mole of GTP (ATP). Like the conversion of pyruvate to acetyl CoA, the citric acid cycle in eukaryotic cells takes place in the matrix of the mitochondria. Unlike glycolysis, the citric acid cycle is a closed loop: The last part of the pathway regenerates the compound used in the first step.



Overall reaction of **tricarboxylic acid (TCA) cycle or Krebs or citric acid cycle**

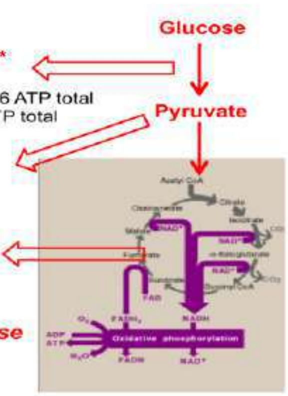
Overall reaction: acetyl CoA + 3 NAD$^+$ + FAD + ADP + P$_i$ + 3 H$_2$O ⟶
2 CO$_2$ + 3 NADH + 3 H$^+$ + FADH$_2$ + ATP + CoA

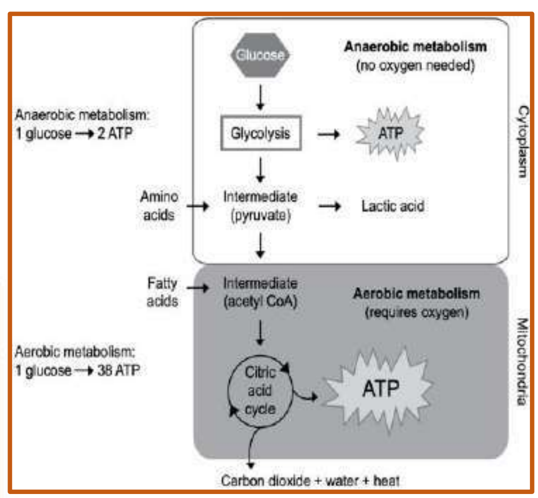
Energy Yield from Glucose Oxidation

- On ETC → 3 ATP per NADH & 2 ATP per FADH$_2$

Pathway	**Yield**
Glycolysis (energy generation stage)	2 ATP 2 NADH → 4-6 ATP**
PDH complex (2X per glucose)	2 NADH → 6 ATP
TCA Cycle (2 turns per glucose)	6 NADH → 18 ATP 2 FADH$_2$ → 4 ATP 2 GTP= 2 ATP
Grand total	*36-38 ATP / glucose*

***if malate-aspartate shuttle 2 NADH (cyto)→2 NADH (mito) →6 ATP total
if glycerol-3P shuttle 2 NADH (cyto) → 2 FADH$_2$ (mito) → 4 ATP total





In the electron transport chain reaction each molecule of NADH will generate 3 ATP and each molecule of FADH2 will generate 2 ATP.

So after complete oxidation of glucose through TCA and elceton transport chain reaction around 38 ATP molecules are generated which is 19 times more energy than that of generated by anaerobic metabolism (which generates 2 ATP molecules). In aerobic respiration all NADH and FADH are converted to ATP. But in anaerobic metabolism NADH produced cant be utilized to generate ATP. So, 2 NADH generated in glycolysis cant be utilized.

Oxidative Phosphorylation/Electron Transport Chain reaction

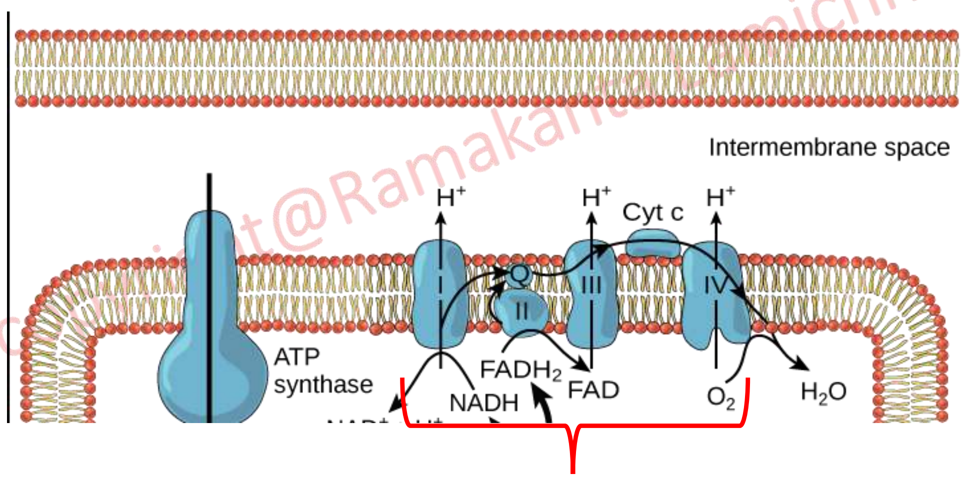
There are two pathways in glucose catabolism—glycolysis and the citric acid cycle—that generate ATP. However, most of the ATP generated during the aerobic catabolism of glucose, is not generated directly from these pathways. Rather, it derives from a process that begins with passing electrons through a series of chemical reactions to a final electron acceptor, oxygen. These reactions take place in specialized protein complexes located in the inner membrane of the mitochondria of eukaryotic organisms and on the inner part of the cell membrane of prokaryotic organisms. The energy of the electrons is harvested and used to generate a electrochemical gradient across the inner mitochondrial membrane. The potential energy of this gradient is used to generate ATP. The entirety of this process is called oxidative phosphorylation.

The electron transport chain reaction is the last component of aerobic respiration and is the only part of glucose metabolism that uses atmospheric oxygen. Oxygen continuously diffuses into plants; in animals, it enters the body through the respiratory system. Electron transport is a series of redox reactions that resemble a relay race in that electrons are passed rapidly from one component to the next, to the endpoint of the chain where the electrons reduce molecular oxygen, producing water. The electron donor molecules are NADH and FADH2. Ultimately the electron transport chain produces 32 molecules of ATP from one molecule of glucose, and also regenerates NAD and FAD for reuse in glycolysis.

Mitochondria: Electron Transport Chain

There are four complexes composed of proteins which also act as hydrogen (proton) pump → I through IV

Electron carriers are labeled as Q (Coenzyme Q) and Cyt C (cytochrome c). The aggregation of these **four complexes**, together **with electron carriers mobile component**, is called the electron transport chain. The **electron transport chain is present in multiple copies** in the inner mitochondrial membrane of eukaryotes. Protein complex pump the proton to create a proton gradient across a membrane.



Electron Transport Chain

Mitochondria: Electron Transport Chain

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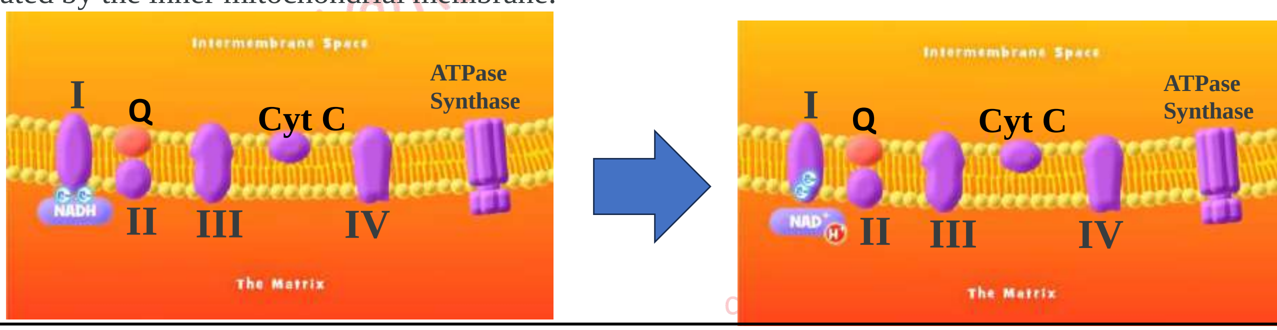
Complex I

To start, two electrons are carried to the first complex by NADH. The enzyme in complex I is NADH dehydrogenase and is a very large protein, containing 45 amino acid chains. Complex I can pump hydrogen ions across the membrane from the matrix into the intermembrane space, and it is in this way that the hydrogen ion gradient is established and maintained between the two compartments separated by the inner mitochondrial membrane.

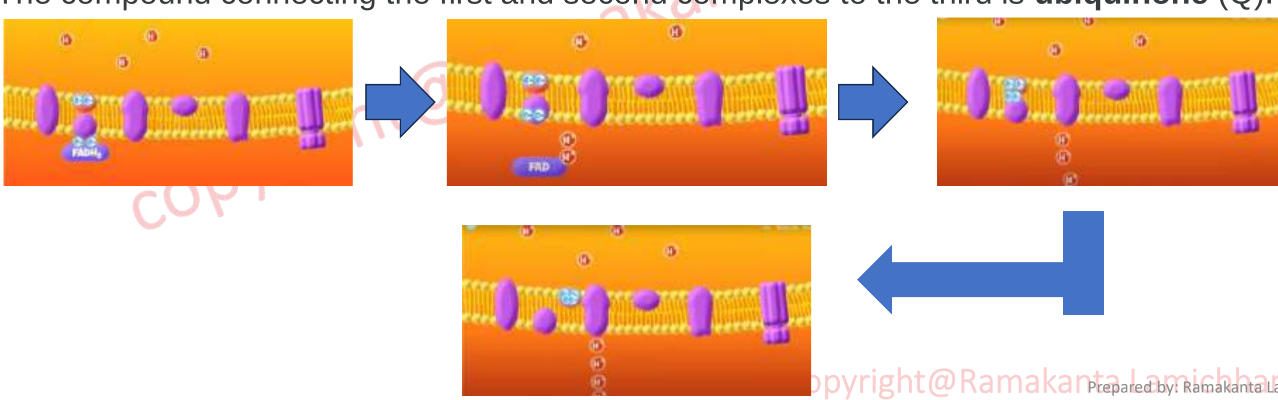


Mitochondria: Electron Transport Chain

Q and Complex II

The Q molecule is lipid soluble and freely moves through the hydrophobic core of the membrane. Once it is reduced, (QH2), ubiquinone delivers its electrons to the next complex in the electron transport chain. Q receives the electrons derived from NADH from complex I and the electrons derived from FADH2 from complex II. Complex II consist of enzyme succinate dehydrogenase. Complex II directly receives $FADH_2$, which does not pass through complex I. Complex II receives electron through FADH2.

The compound connecting the first and second complexes to the third is **ubiquinone** (Q).



Mitochondria: Electron Transport Chain

Complex III

The third complex is composed of a protein called cytochrome b and cytochrome c proteins; this complex is also called cytochrome oxidoreductase. Cytochrome proteins have a prosthetic group of heme. The heme molecule is similar to the heme in hemoglobin, but it carries electrons, not oxygen. As a result, the iron ion at its core is reduced and oxidized as it passes the electrons, fluctuating between different oxidation states: Fe^{++} (reduced) and Fe^{+++} (oxidized). Complex III pumps protons through the membrane and passes its electrons to cytochrome c for transport to the fourth complex of proteins and enzymes (cytochrome c is the acceptor of electrons from Q; however, whereas **Q carries pairs of electrons, cytochrome c can accept only one at a time**).

Complex IV

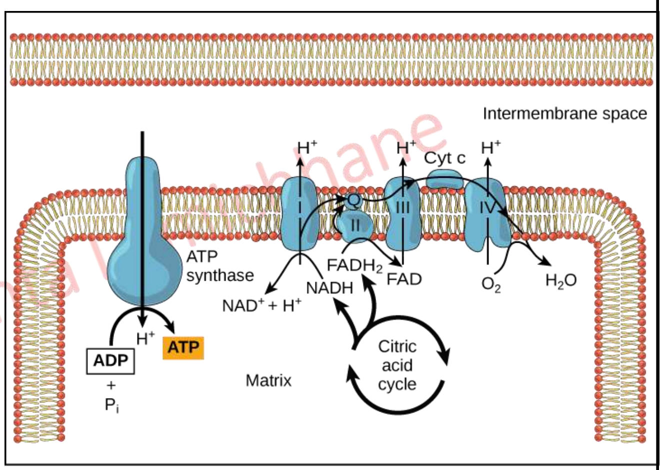
The fourth complex is composed of cytochrome proteins c, a, and a_3. This complex contains two heme groups (one in each of the two cytochromes→ a, and a_3) and three copper ions. The cytochromes hold an oxygen molecule very tightly between the iron and copper ions until the oxygen is completely reduced. The reduced oxygen then picks up two hydrogen ions from the surrounding medium to make water (H_2O). The removal of the hydrogen ions from the system contributes to the ion gradient used in the process of chemiosmosis.

Mitochondria: Electron Transport Chain

Chemiosmosis

The uneven distribution/aggregation of H^+ ions (higher outside the inner membrane) occurs across the membrane due to protein pumps. This result in both concentration and electrical gradients (electrochemical gradient). In chemiosmosis, diffusion of hydrogen ion occurs across membrane to balance the concentration.

If the membrane were open to diffusion by the hydrogen ions, the ions would tend to diffuse back across into the matrix, driven by their electrochemical gradient. Recall that many ions cannot diffuse through the nonpolar regions of phospholipid membranes without the aid of ion channels. Similarly, hydrogen ions in the matrix space can only pass through the inner mitochondrial membrane through an integral membrane protein called ATP synthase.



This complex protein acts as a tiny generator, turned by the force of the hydrogen ions diffusing through it, down their electrochemical gradient. The turning of parts of this molecular machine facilitates the addition of a phosphate to ADP, forming ATP, using the potential energy of the hydrogen ion gradient

Chloroplast: Ultrastructure, chloroplast metabolism, photosynthesis electron transport

Chloroplasts are found in all green plants and algae. They are the food producers of plants. These are found in mesophyll cells located in the leaves of the plants. They contain a high concentration of chlorophyll that traps sunlight. This **cell organelle** is not present in animal cells. Higher plants have lens-shaped chloroplast. The sizes are: 2 – 4 um wide and 5 -10 um long. The number ranges from 20 – 40 per cell. They are giant organelles compared to other organelles in the cell and they are as large as an entire mammalian red blood cell.

Chloroplasts are reported to be found in all green parts of the plants like the leaves, young branches, stems, sepals, unripe fruits and also in roots of certain plants like Taeniophyllum (orchid), Tinospora etc.

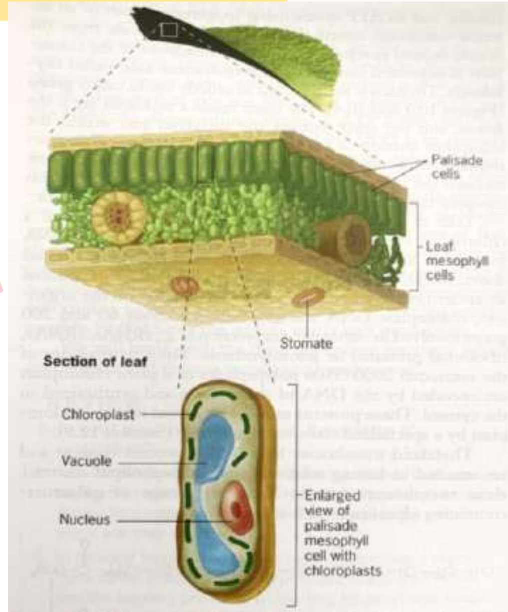
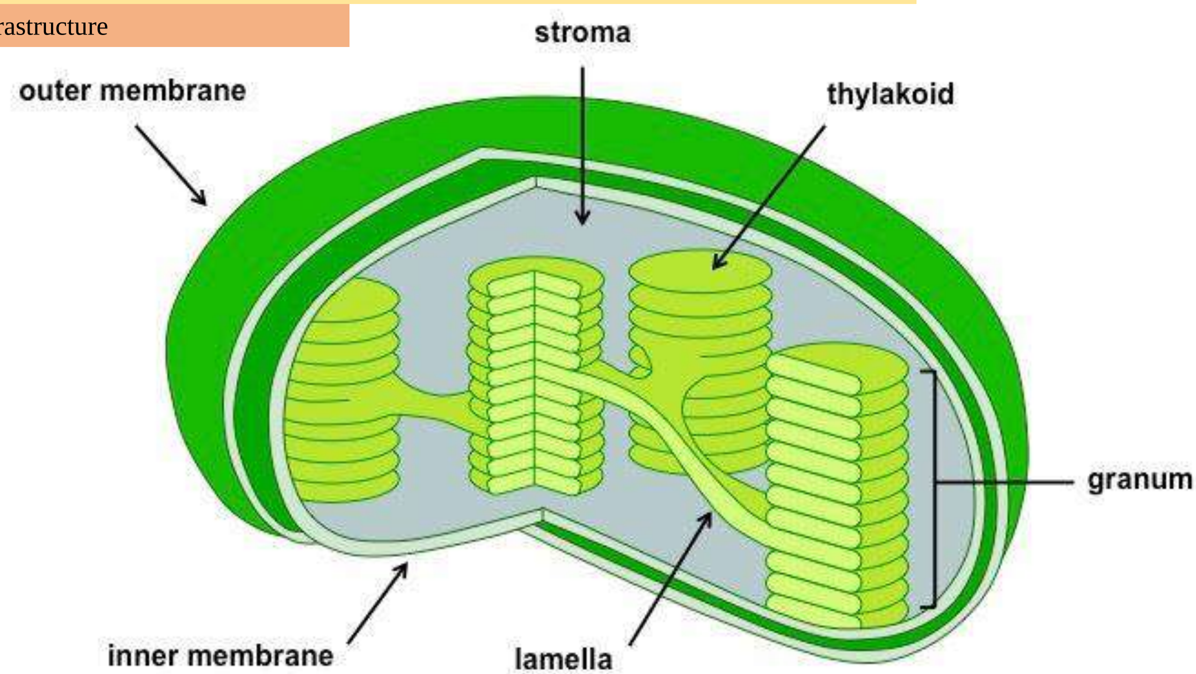


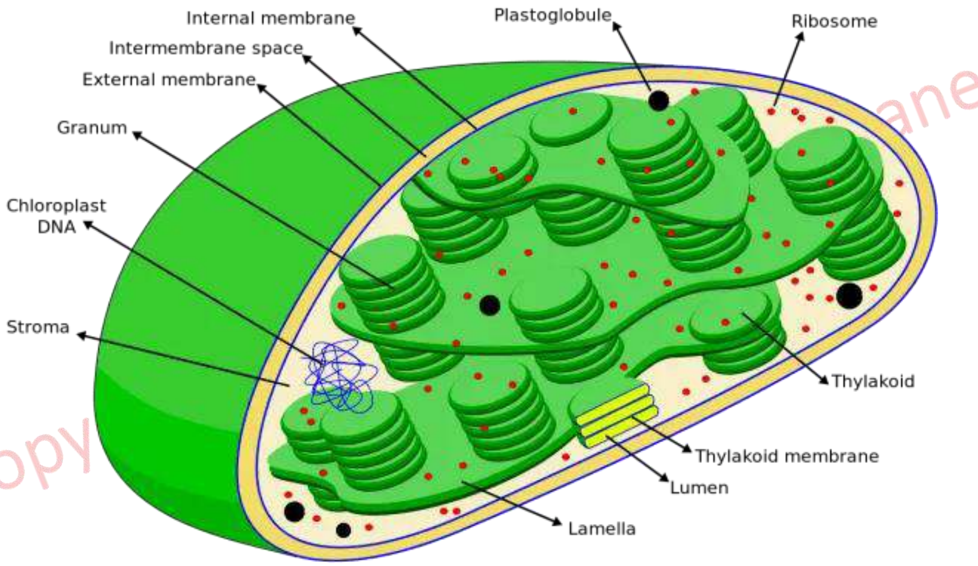
Figure 10.2 **The functional organization of a leaf.** The section of the leaf shows several layers of cells that contain chloroplasts distributed within their cytoplasm. These chloroplasts carry out photosynthesis, providing raw materials and chemical energy for the entire plant.

Chloroplast: Ultrastructure, chloroplast metabolism, photosynthesis electron transport

2. Cell organelles & functions

Chloroplast: Ultrastructure





Chloroplast: Ultrastructure

Outer covering of chloroplast consists of two membrane separated by a narrow space. Outer membrane contains relatively large channels which more permeable than the inner channel. The inner membrane of the envelope is highly impermeable, and allow the movement of substance with the aid of variety of transporters.

Plastoglobule: lipoprotein particles

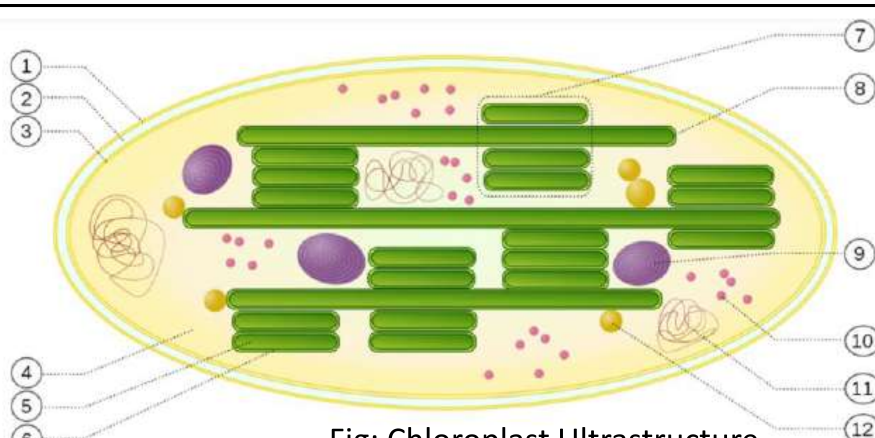
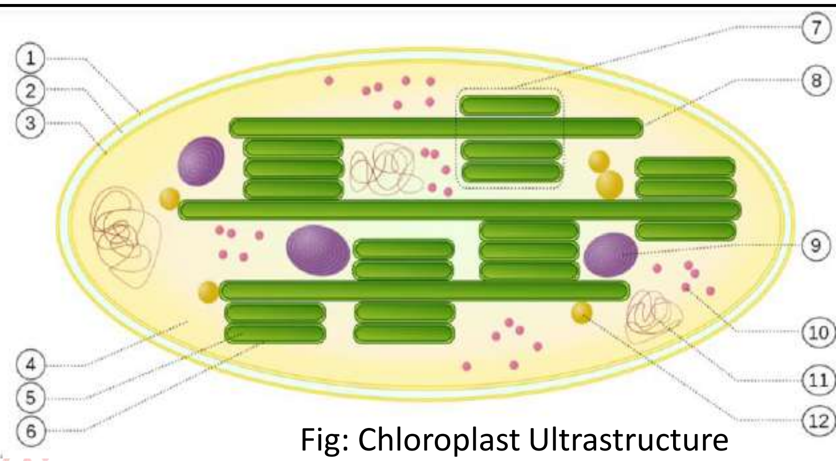


Fig: Chloroplast Ultrastructure

1. Outer membrane
2. Intermediate membrane
3. Inner membrane
4. Stroma
5. Thylakoid lumen
6. Thylakoid membrane
7. Grana
8. Thylakoid
9. Starch
10. Ribosomes
11. Chloroplast DNA
12. Plastoglobule

Chloroplast: Ultrastructure

The main component of chloroplast responsible for photosynthesis are the light-absorbing pigments (chlorophyll), a complex chain of reaction carriers and an ATP-synthesizing apparatus are present in an internal membrane system called thylakoids. It is physically separate form the double-layred envelop. Thylakoids are flattened membranous sacs, which are arranged in orderly stacks called grana.



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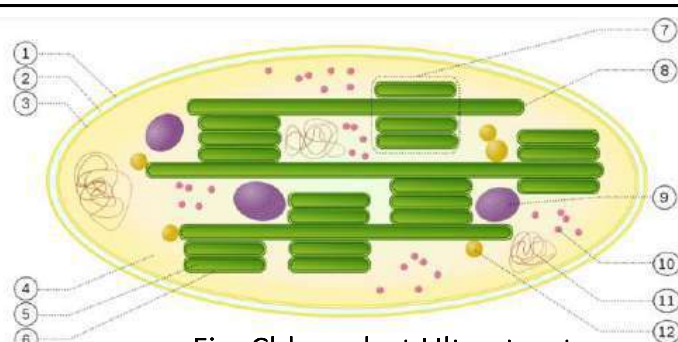
12. Plastoglobule

Chloroplast: Ultrastructure

The space inside a thylakoid sac is the lumen and the space outside the thylakoid and within the chloroplast envelope is the stroma. The stroma contains the enzymes responsible for carbohydrate synthesis. The photosynthetic pigments like chlorophylls and carotenoids are present in the thylakoid membrane.



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11. Chloroplast DNA
12. Plastoglobule

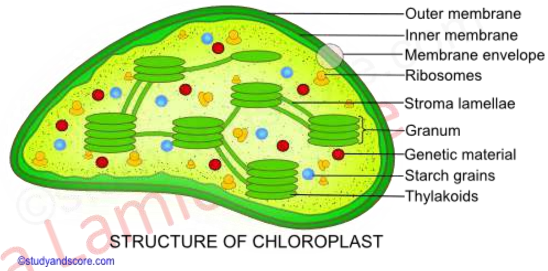
Like the matrix of a mitochondrion, the stroma of a chloroplast contains small, double-stranded, circular DNA molecules and prokaryotic-like ribosomes. Chloroplast DNA contain genes for formation of tRNA, rRNA, ribosomal proteins and photosynthetic proteins.

Ultrastructure of chloroplasts

(Extra information)

A chloroplast comprises of three main components namely Envelope, Stroma and Thylakoids.

Envelope: The chloroplast is bound by an envelope which is made up of two unit membranes. These unit membranes are separated by periplastidal space of 10nm. Exchange of molecules between chloroplast and cytosol occurs across this double membrane envelope. The inner envelope acts like a control gate and regulator, controlling the flow of necessary material and particles into, and out of, the chloroplast. The small, simple, and important molecules like water (H_2O), carbon dioxide (CO_2), and oxygen (O_2) can pass through this membrane. Isolated membranes of envelope do not have chlorophyll pigment and cytochromes. The isolated membranes appear yellow due to the presence of small amounts of carotenoids.



Stroma: The inner space of chloroplast is filled with a colorless matrix known as stroma. Stroma fills most of the volume of the chloroplast. Stroma surrounds the thylakoids and contains almost 50% of the proteins of the chloroplast. The proteins present in the stroma are soluble in nature. A large number of lipoprotein membranes extend from one end to the other called as stroma lamellae.

Stroma is also found to contain 70s ribosomes and DNA molecules. These ribosomes and DNA molecules synthesize some of the structural proteins of the chloroplast. Carbon fixation, synthesis of sugars, starch and fatty acids also takes place inside the stroma.

Thylakoids: Thylakoids consists of closed and flattened vesicles. These vesicles are arranged as a membranous network. The outer surface of the thylakoid is in contact with the stroma and the inner surface encloses an intra-thylakoid space. Thylakoids are found to be stacked one on other like the pile of coins. This stack of thylakoids is called as grana. The number of grana in the matrix of a chloroplast may be 40 to 80.

The thylakoids present in the granum are called as granum thylakoids. The space within the thylakoid is called as lumen. The photosynthetic pigments like chlorophylls and carotenoids are present in the thylakoid membrane.

The presence of DNA in the chloroplasts helps in self-duplication. Hence chloroplasts are also called as semi-autonomous organelles.

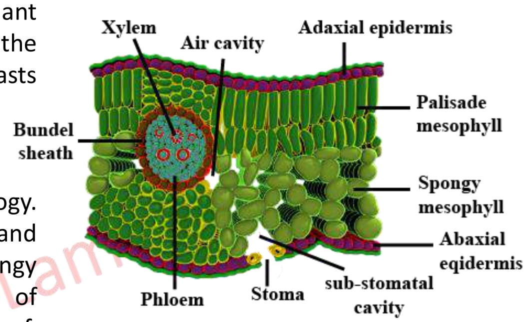
Present-day plant chloroplasts belong to the plastid organelle family that include specialized organelles such as amyloplasts (starch storage), chromoplasts (pigmented plastids), leucoplasts (storage), elaioplasts (lipid storage), gerontoplasts (senescing plastids) and etioplasts (developing plastids) (Rolland et al., 2018). Only the green chloroplasts are photosynthetically competent. The structure of mature chloroplasts is defined by three membrane systems: the outer and inner envelope membranes and the thylakoid membrane system (Staehelin, 1986). The organization of these three membranes defines three separate aqueous compartments

Chloroplasts are reported to be found in all green parts of the plants like the leaves, young branches, stems, sepals, unripe fruits and also in roots of certain plants like Taeniophyllum (orchid), Tinospora etc.

Chloroplasts hold several world records in biology: they host the most abundant soluble protein (ribulose 1,5 bisphosphate carboxylase/oxygenase, Rubisco; see Box 1 for abbreviations), the most abundant membrane protein (light-harvesting complex II, LHCII), the most abundant pigments (chlorophyll, Chl), and the most abundant lipid (monogalactosyldiacylglycerol, MGDG). The uniqueness of chloroplasts is based, rather, on their capability of making use of solar radiation to generate metabolic energy equivalents and, at the same time, to channel these equivalents with high productivity into a battery of anabolic reactions in the chloroplast stroma or to export them to the cytoplasm and beyond (e.g. sugar export via phloem in plants).

(Extra information)

The chloroplasts are distributed homogeneously in the cytoplasm of the plant cells. But in certain plant cells they may remain concentrated around the nucleus. They have definite orientation in the cell cytoplasm. As chloroplasts are motile organelles they show both passive and active movements.



Chloroplast, leaf, location, mesophyll, chloroplast, distribution, morphology. The chloroplasts are located in the mesophyll region, between the upper and the lower epidermis. It consists of several layers of loosely arranged spongy parenchyma cells with intercellular spaces. Due to the presence of chloroplasts, mesophyll region is the chief photosynthetic tissue of the leaf. The number of chloroplast in a single mesophyll cell ranges from 1 to 50. This number may vary from cell to cell depending on plant species, age, and health of the cell.

(Extra information)

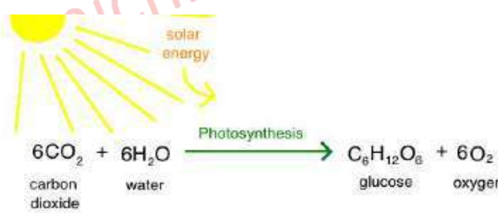
Chloroplast: Photosynthetic Metabolism

2. Cell organelles & functions

From which molecule between CO_2 and H_2O, will generate molecular oxygen in photosynthesis?

Photosynthesis is **the process by which plants use sunlight, water, and carbon dioxide to create oxygen and energy in the form of sugar**. Carbon from carbon dioxide—inorganic carbon—can be incorporated into organic molecules; this process is called **carbon fixation**. The carbon that's fixed and incorporated into sugars during photosynthesis can be used to build other types of organic molecules needed by cells.

In this scheme, each molecule of oxygen is derived not form CO_2, but from the breakdown of two molecules of H_2O, a process driven by the absorption of light. This role of water was established by scientist in 1941. They carried out two experiments on suspension of green algae using a specially labeled isotope of oxygen, ^{18}O, as a replacement for the common ^{16}O and unlabeled water. In the first experiment they used $C[^{18}O_2]$ and unlabeled water. In second experiment sample was exposed to unlabeled carbon dioxide and labeled $H_2[^{18}O]$



+ATP

Now the question was which experiment releases labeled $^{18}O_2$?
The algae given labeled water produced labeled oxygen demonstrating that O_2 produced during photosynthesis derived from H_2O. The algae given labeled carbon dioxide produced unlabeled oxygen, confirming that O_2 was not being produced by a chemical splitting of CO_2.

Chloroplast: Photosynthetic Metabolism

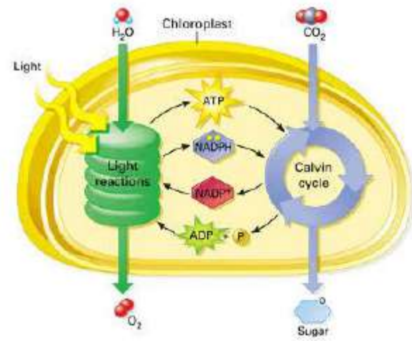
Functions of Chloroplasts

The chloroplasts are chiefly concerned with assimilation of food materials by photosynthesis. This process of photosynthesis occurs in two stages or phases:

I. Light dependent reaction (taking place in the thylakoid membrane and thylakoid space)

The first part of the photosynthetic process, the light-dependent reaction, takes place in the thylakoids membrane. The thylakoids are stacked and are connected to one another by the stroma lamellae. The outer surface of thylakoid is called the thylakoid membrane. The pigments and ATP synthase molecules which help in photosynthesis are present on the thylakoid membrane and so this membrane is also called as photosynthetic membrane.

The captured solar energy knocks out an electron which provides the energy to other molecules to start the process of photosynthesis and produce ATP and NADPH. the light-dependent phase which needs the energy from light to convert it to chemical energy in the form of ATP and NADPH molecules.



Functions of Chloroplasts

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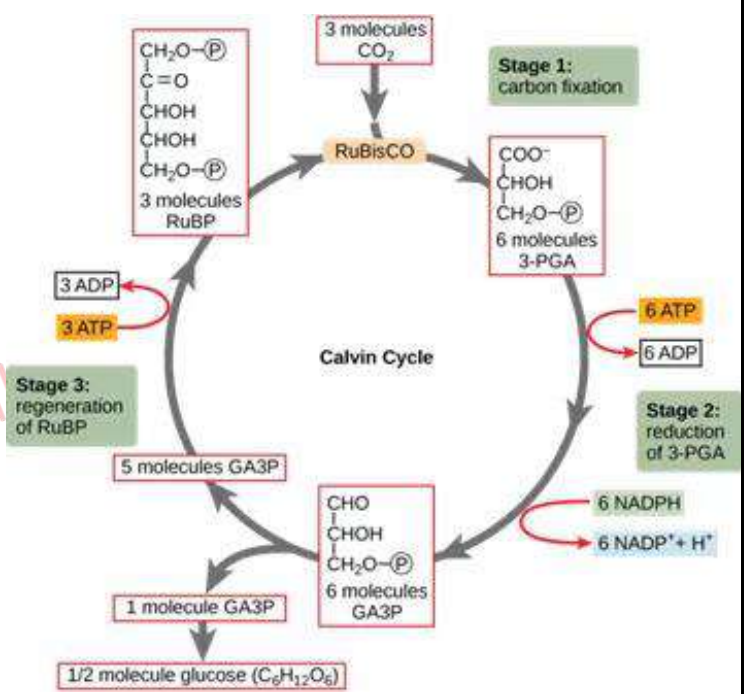
II. Light independent reaction (taking place in the stroma)

This is the second stage which is also called dark reaction. The second phase is known as the Calvin cycle, takes place in the stroma. Here, the carbohydrates are synthesized from the carbon dioxide using the energy stored in the ATP and NADPH molecules that are produced in the light-dependent reactions, i.e. the first phase of photosynthesis.



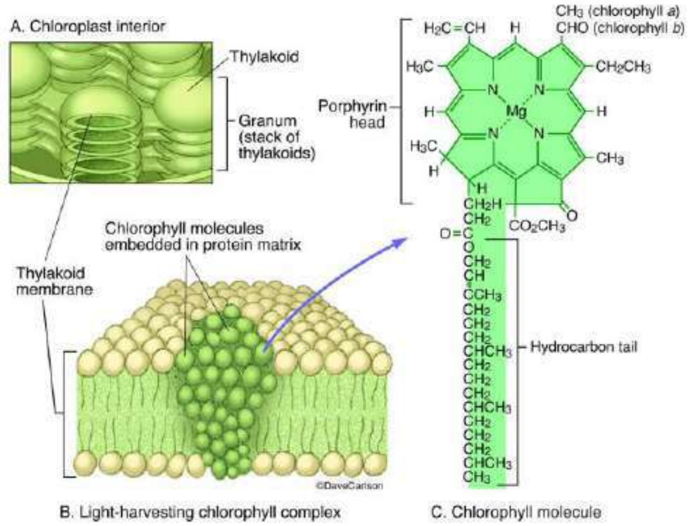
Photosynthetic pigments

Pigments are compounds that appear colored because they only absorb light of particular wavelength(s) within the visible spectrum. Leaves are green because their chloroplast contain large quantities of the pigment called chlorophyll, which absorbs most strongly in the blue and red, leaving the intermediate green wavelengths to be reflected to our eyes.

Each molecule consists of two parts:

1. A porphyrin ring (head) that function in light absorption
2. A hydrocarbon chain (phytol chain) that keeps the chlorophyll embedded in the photosynthetic membrane

Chlorophyll a is present in all oxygen-producing photosynthetic organism. In higher plants and algae, in addition to chlorophyll a, **chlorophyll b is** also present. The porphyrins are also present in haemoglobin and myoglobin which contain iron in the centre, but in the porphyrins of chlorophyll, magnesium atom is present at the centre. The alternating single and double bonds along the edge of the porphyrin ring delocalize electrons, which forms a cloud around the ring. System of this type are said to be conjugated and are strong absorber of visible light. The absorbed energy cause a redistribution of the electron density of the molecule, which in turn favours loss of an electron to the suitable acceptor.



Photosynthesis electron transport

Below figure is a diagrammatic representation of the thylakoid membrane. As we can see, embedded in the membrane are the photosystems (I and II) that absorb light energy. Also, found within the membrane are electron carrier proteins, proton pumps, and the enzyme ATP synthase. Together, these structures make up the components of the electron transport chain.

In the first series of reactions, light falls on photosystem II. The chlorophyll pigments within photosystem II absorb this energy, and the electrons within them are excited. The electrons move to a higher energy level and are transferred to the primary electron acceptor within the photosystem.

As these electrons move to an electron acceptor, they need to be replaced. The absorption of light energy by photosystem II not only excites electrons to higher energy levels but is used to split a molecule of water into an oxygen atom and hydrogen ions, releasing electrons in the process. This reaction is known as photolysis, which means “breakdown using light.” This splitting of water releases electrons that are used to replace the excited electrons that have been transferred to the primary electron acceptor. It also releases two hydrogen (H+) ions into the interior of the thylakoid.

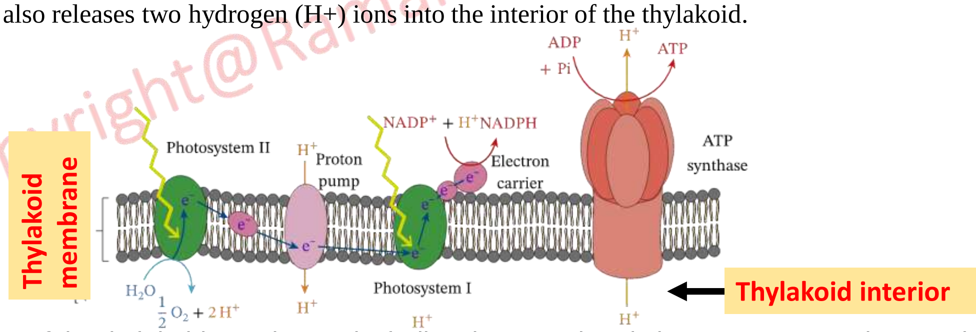


Figure 2: A diagram of the thylakoid membrane, including the associated photosystems, carrier proteins, and ATP synthase

Photosynthesis electron transport

The electrons that have left photosystem II now move down the electron transport chain, releasing energy as they are passed between each component. This energy is used to actively transport more hydrogen ions from the stroma of the chloroplast into the thylakoid interior. As the electrons reach photosystem I, they are excited again, as the chlorophyll pigments in this photosystem absorb light energy. The electrons move to a higher energy level and are transferred to the primary electron acceptor in photosystem I. From here, they are passed to more protein carriers and eventually to the enzyme $NADP^+$ Reductase.

$NADP^+$ is a coenzyme found in the chloroplast. It acts as a universal electron acceptor. The electrons that have moved down the electron transport chain and have reached $NADP^+$ reductase are used, along with a hydrogen ion, to reduce $NADP^+$ to NADPH. This is an important step as NADPH is a key coenzyme used in the next stage of photosynthesis, the light-independent stage.

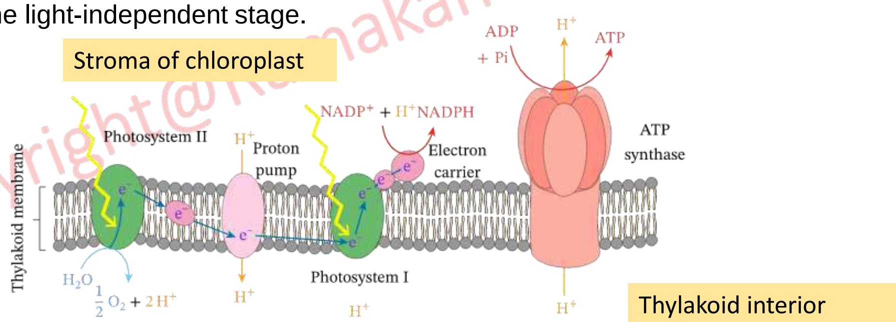


Figure 2: A diagram of the thylakoid membrane, including the associated photosystems, carrier proteins, and ATP synthase.

Photosynthesis in chloroplast and respiration in mitochondria

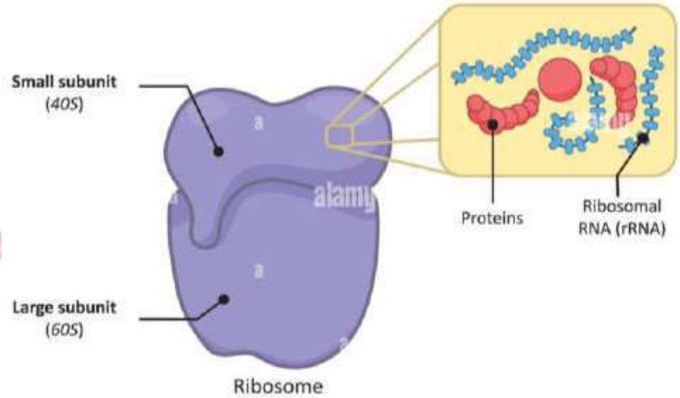
Respiration in mitochondria reduces oxygen to water, but in photosynthesis in chloroplast oxidizes water to oxygen. Both processes release energy, but the latter process must require energy i.e. sunlight without which photosynthesis cannot occur.

Photosynthesis and cellular respiration both involve a series of **redox** reactions (reactions involving electron transfers). In cellular respiration, electrons flow from glucose to oxygen, forming water and releasing energy. In photosynthesis, they go in the opposite direction, splitting water to produce oxygen and additional production of glucose—an energy-requiring process powered by light. Like cellular respiration, photosynthesis also uses an electron transport chain to make H^+ concentration gradient, which drives ATP synthesis by chemiosmosis.

Write differences and similarities (as many as possible) between **photosynthesis in chloroplast** and **respiration in mitochondria**.
(Assignment to all students: submit in papers) Submission date: within one week

Ribosome: Ultrastructure, ribosome subunits, polysomes, protein synthesis, SRP-ribosome complex

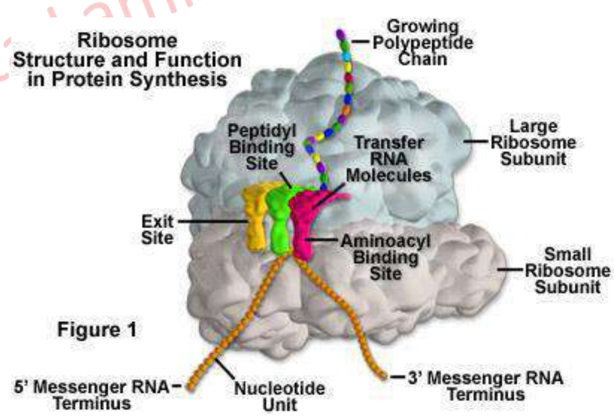
Ribosomes are tiny spheroidal dense particles (of 150 to 200 Å diameters). They are sites of **protein synthesis**. It plays an important role in decoding the genetic message reserved in the genome into protein. They are structures containing RNA (60 %) and proteins (40%). The ribosomes occur in cells, both prokaryotic and eukaryotic cells. In prokaryotic cells, the ribosomes often occur freely in the cytoplasm. In eukaryotic cells, the ribosomes either occur freely in the cytoplasm or remain attached to the outer surface of the membrane of the endoplasmic reticulum or in the organelles. The location of the ribosomes in a cell determines what kind of protein it makes. If the ribosomes are floating freely throughout the cell, it will make proteins that will be utilized within the cell itself (cell's own internal use). When ribosomes are attached to the endoplasmic reticulum, it is referred to as rough endoplasmic reticulum or rough ER. Proteins made on the rough ER are used for usage inside the cell or outside the cell.



Ribosome: Ultrastructure, ribosome subunits, polysomes, protein synthesis, SRP-ribosome complex

The number of ribosomes in a cell depends on the activity of the cell. On average in a mammalian cell, there can be about 10 million ribosomes. However, though they are generally described as organelles, it is important to note that ribosomes are not bound by a membrane and are much smaller than other organelles. They require the use of an electron microscope to be visually detected.

In eukaryotes, the rRNA in ribosomes is organized into four strands, and in prokaryotes, three strands. Eukaryote ribosomes are produced and assembled in the nucleolus. Ribosomal proteins enter the nucleolus and combine with the four rRNA strands to create the two ribosomal subunits (one small and one large) that will make up the completed ribosome. The proteins in a ribosome help fill in structural gaps and enhance protein synthesis. The ribosome units leave the nucleus through the nuclear pores and unite once in the cytoplasm for the purpose of protein synthesis. When protein production is not being carried out, the two subunits of a ribosome are separated.

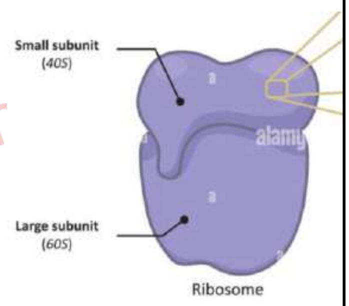


Ribosome: Ultrastructure,

Structure of Ribosomes

Each ribosome is divided into two subunits:

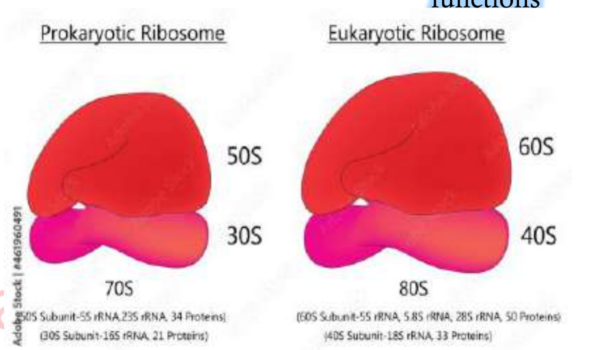
1. **A smaller subunit** which binds to a larger subunit and the mRNA. The smaller subunit mediates the correct interactions between three consecutive, in-frame base pairs, called codons and the complementary triplet called the anti-codon on the transfer RNA (tRNA).
2. **A larger subunit** which binds to the tRNA, the amino acids, and the smaller subunit. The larger subunit, which is approximately twice the size of the smaller, contains the peptidyl-transferase center (PTC), which catalyzes the condensation reaction that mediates the formation of the peptide bond between amino acids in the growing polypeptide. The ribosomes share a core structure that is similar to all ribosomes despite differences in its size.



2. Cell organelles & functions

Structure of Ribosomes

Prokaryotic ribosomes are bacterial ribosomes that are small (70S) while eukaryotic ribosomes are large ribosomes (80S). The simplest ribosome is from bacteria and is comprised of a 30S and 50S subunit which together form the 70S ribosome. Contrastingly, yeast ribosomes are comprised of a 40S and 60S subunit, which assemble to form the 80S structure. S is an abbreviation for Svedberg's unit and it represents the sedimentation coefficient of a molecule measured by centrifuge. It refers to the rate at which molecules sediment in centrifuges run at very high speed.



Ribosomes in mitochondria and chloroplast

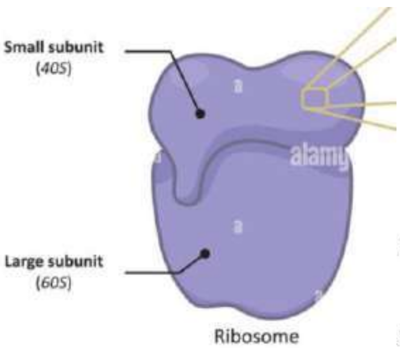
In addition to the most familiar cellular locations of ribosomes, the organelles can also be found inside mitochondria and the chloroplasts of plants. These ribosomes notably differ in size and makeup than other ribosomes found in eukaryotic cells, and are more akin to those present in bacteria and blue-green algae cells.

Source of Ribosome	Sedimentation Coefficient	Subunits
Bacteria	70S	30S and 50S
Eukaryotic cytosol	80S	40S and 60S
Mammalian Mitochondria	55S	28S and 39S

The similarity of mitochondrial and chloroplast ribosomes to prokaryotic ribosomes is generally considered strong supportive evidence that mitochondria and chloroplasts evolved from ancestral prokaryotes.

Ribosome: Ultrastructure,

The two subunits fit together and work as one to translate the mRNA into a polypeptide chain during protein synthesis. The two subunits assemble to translate mRNA and disassemble when translation is complete. So, the existence of ribosomes is temporary, after the synthesis of polypeptide the two sub-units separate and are reused or broken up. During protein synthesis, when multiple ribosomes are attached to the same mRNA strand, this structure is known as polysome.



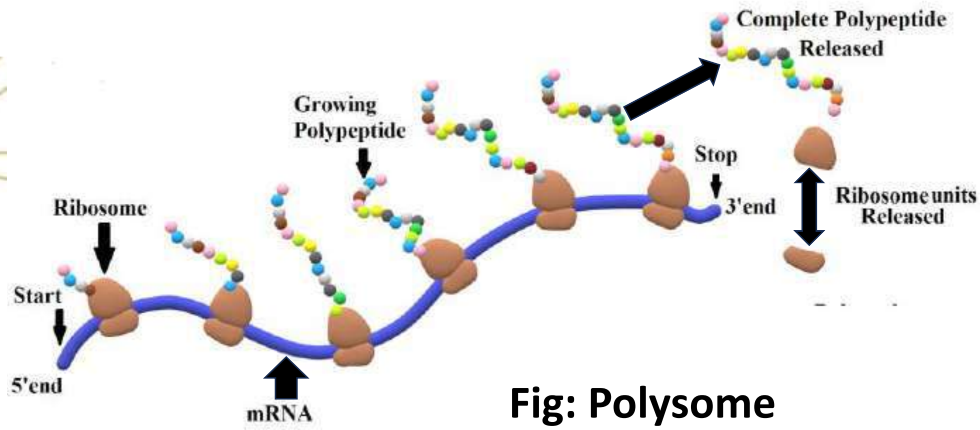
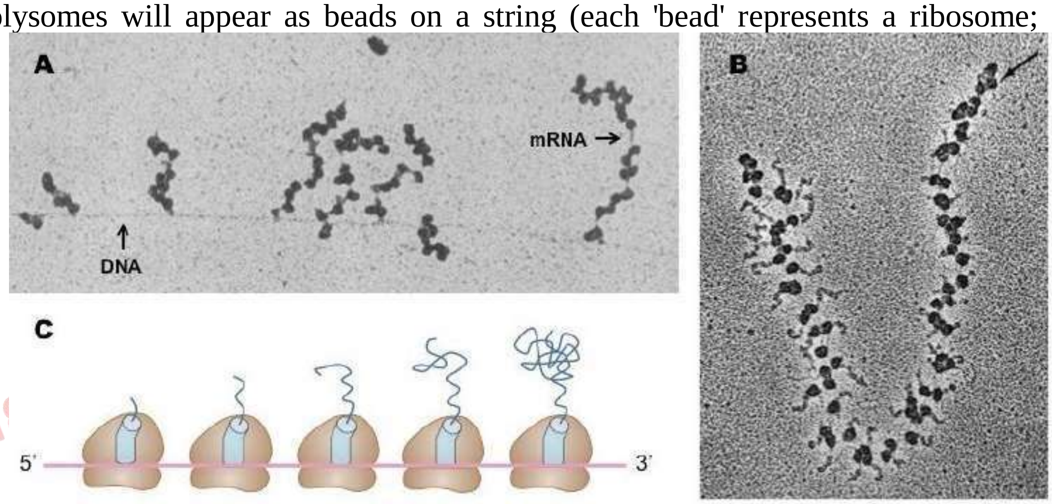


Fig: Polysome

Ribosome: Polysomes

A **polysome** (or a polyribosome) is a group of two or more ribosomes translating an mRNA sequence simultaneously. The polysomes will appear as beads on a string (each 'bead' represents a ribosome; the 'string' is the mRNA strand).

In prokaryotes, the polysomes may form while the mRNA is still being transcribed from the DNA template. Ribosomes located at the 3'-end of the polysome cluster will have longer polypeptide chains than those at the 5'-end.



A: Polysome formation on RNA strands being transcribed from a DNA template (prokaryote)
B: Polysome formation on a eukaryotic mRNA transcript (arrow indicates 5'-end of transcript)
C: Diagrammatic representation of polysomes (polypeptide chain gets longer as ribosome moves 5' → 3')

Ribosome: Polysomes

Benefits of polysomes:

1. It requires less mRNA to be transcribed in order to make the quantity of protein needed. You rarely ever need only one protein molecule to be made. A cell might require a large influx of a certain protein, but it would be rather wasteful to have to make one mRNA for every protein molecule you wanted to make. Allowing one transcript to be read by many ribosomes saves on resources since all those nucleic acids used to make mRNA have to made at some point and that costs ATP.
2. It saves time. Multiple ribosomes making multiple proteins at the same time is simply faster than multiple ribosomes making multiple proteins one after the other. Plus, if you need this protein immediately, its better if you don't have to wait for more rounds of transcription than is absolutely necessary.

Assignment: PowerPoint presentation
Amino acids
Roll No……

Content:
Basic structure
Reaction for the formation of peptide bond
Types: essential and non essential
Figure and process

Ribosome: RNA translation and protein synthesis

There are three stage of protein synthesis in ribosome. In other words, the whole translation process can be divided into three stages:

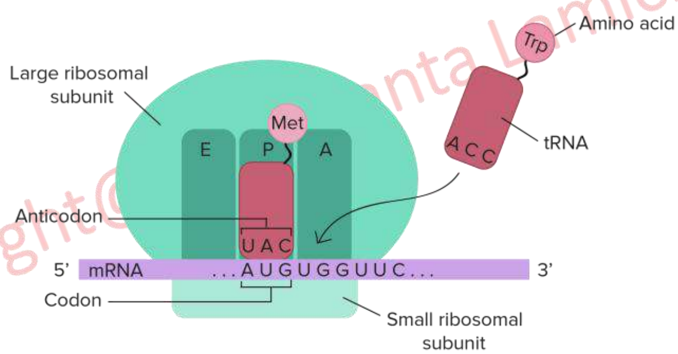
❖**Initiation** (starting off),
❖**Elongation** (adding on to the protein chain), and
❖**Termination** (finishing up).

Initiation

Here the formation of initiation complex occurs which will start the process of protein synthesis. The ribosome, mRNA and tRNA come close to form the initiation complex. The ribosome assembles around the mRNA to be read and the first tRNA carrying the amino acid methionine, bind to its match i.e. the start codon, AUG. This setup, called the initiation complex, is needed in order for translation to get started. If we see in detail, the small ribosomal subunit (30S) binds first to the mRNA sequence. Then the tRNA with methionine binds to the start codon of mRNa (AUG). Then finally the large ribosomal subunit (50S) binds to form a complete initiation complex.

Ribosome: RNA translation and protein synthesis

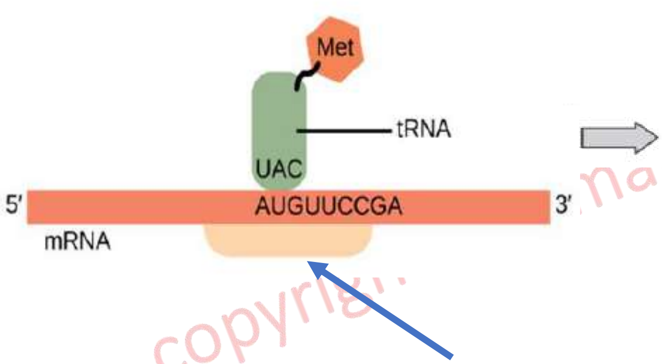
Four binding sites are located on the ribosome, one for mRNA and three for tRNA. The three tRNA sites are labeled P, A, and E. The A site (**acceptor site**), binds to the aminoacyl tRNA, which holds the new amino acid to be added to the polypeptide chain. The P site, called the **peptidyl site**, binds to the tRNA holding the growing polypeptide chain of amino acids. E (**exit**), which mobilizes the emerging tRNA which has been stripped of its amino acid (deacylated-tRNA).

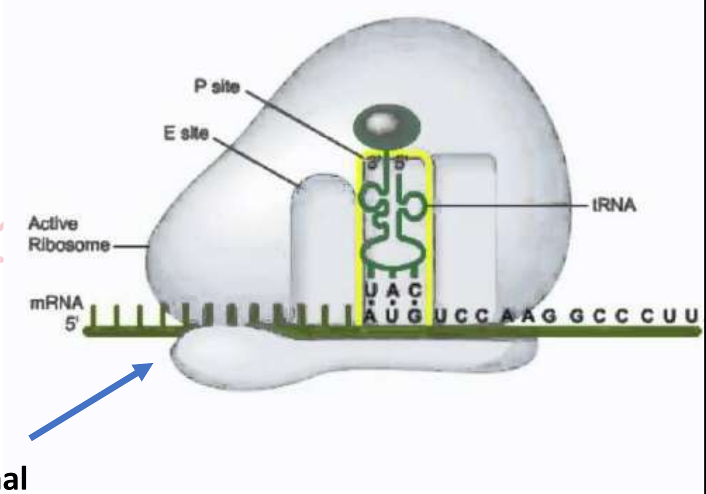


Assignment: short presentation: mRNA and tRNA. (Roll NO....)

Ribosome: RNA translation and protein synthesis

Fig: Initiation Complex





Smaller Ribosomal subunit

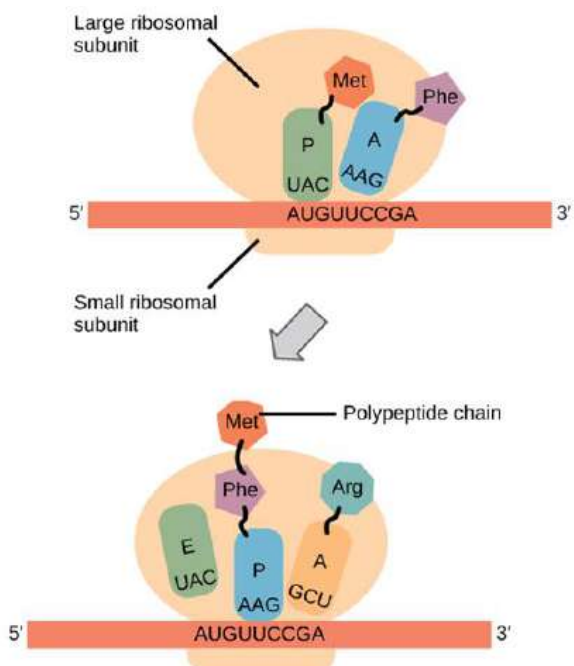
Ribosome: RNA translation and protein synthesis

Elongation

It is the process of extending the chain. Elongation is the stage where the amino acid chain gets longer. In elongation, the mRNA is read one codon at a time, and the amino acid matching each codon is added to a growing protein chain. As the ribosome moves along the mRNA (towards 3' direction), each time a new codon is exposed at the A site. A matching tRNA binds to the codon. The existing amino acid chain (polypeptide) is linked onto the amino acid of the tRNA via a chemical reaction. The mRNA is shifted one codon over in the ribosome, exposing a new codon for reading. During elongation, tRNAs move through the A, P, and E sites of the ribosome, as shown above. This process repeats many times as new codons are read and new amino acids are added to the chain.

Ribosome: RNA translation and protein synthesis

Elongation



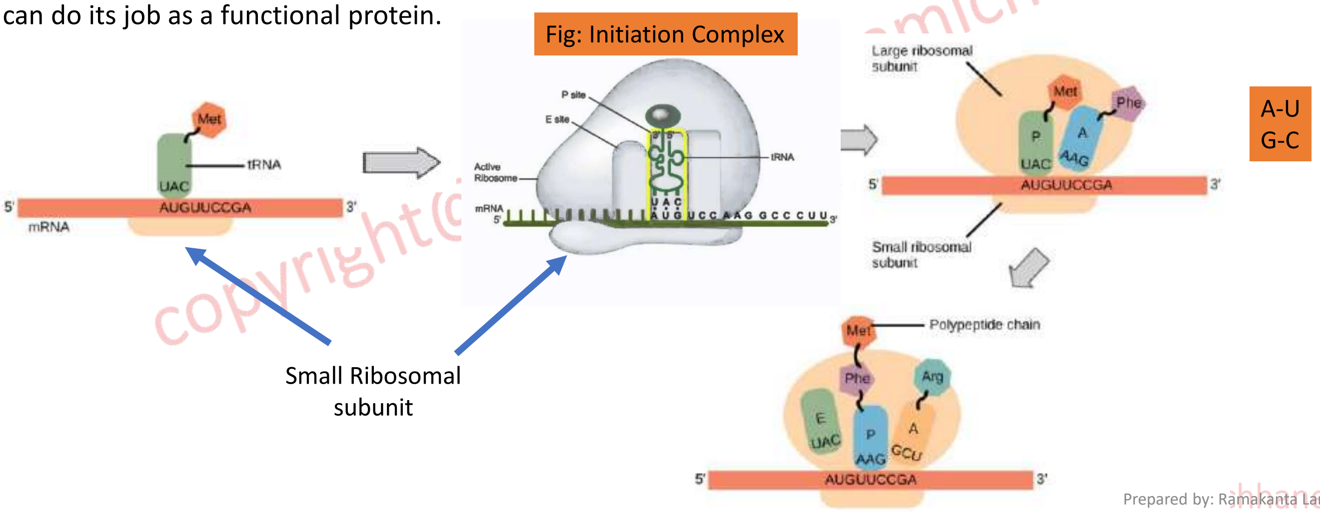
A-U
G-C

Ribosome: RNA translation and protein synthesis

Termination (Finishing up)

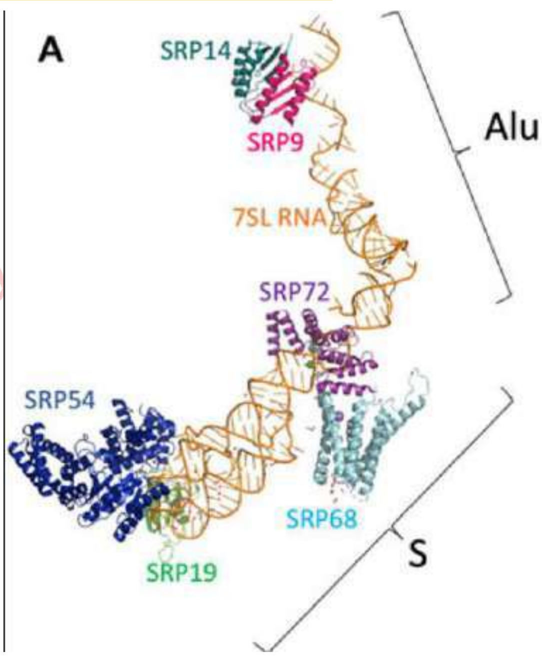
Termination is the stage in which the finished polypeptide chain is released. It begins when a stop codon (UAG, UAA, or UGA) enters the ribosome, and it recognizes the translation is complete. Then there will be the triggering of a series of events that separate the chain from its tRNA and allow it to drift out of the ribosome.

After termination, the polypeptide may still need to fold into the right 3D shape, undergo processing (such as the removal of amino acids), get shipped to the right place in the cell, or combine with other polypeptides before it can do its job as a functional protein.



Signal Recognition Particle (SRP)

Signal Recognition Particle (SRP) is the complex of RNA and protein. SRP is a ribonucleoprotein consisting of six protein subunits arranged on a long noncoding RNA. So, it is also called as ribonucleoprotein complex. The figure represents a model of SRP in which The SRP protein subunits are assembled on RNA. SL RNA (in orange color) is bound to different types of domain proteins like SRP9 (magenta), SRP14 (dark green), SRP72 (purple), SRP68 (light blue), SRP19 (light green), and SRP54 (dark blue).

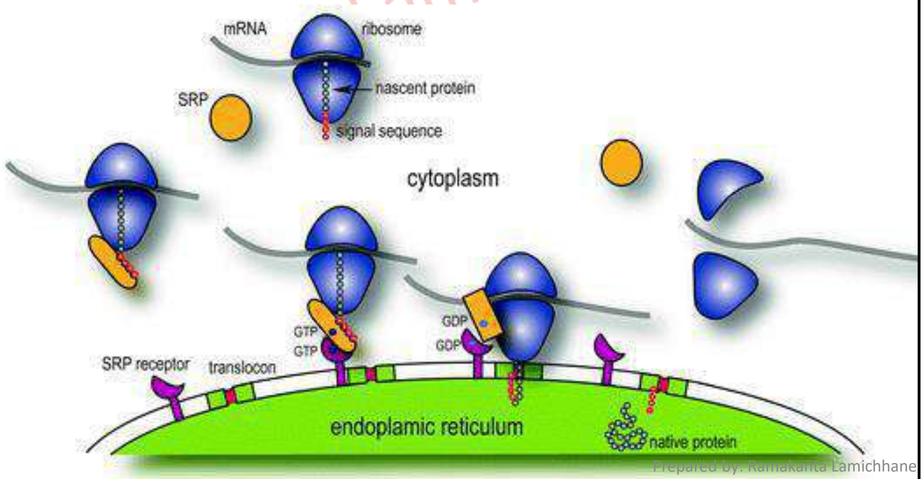


Role of SRP

Targeting secretory and membrane protein to the endoplasmic reticulum

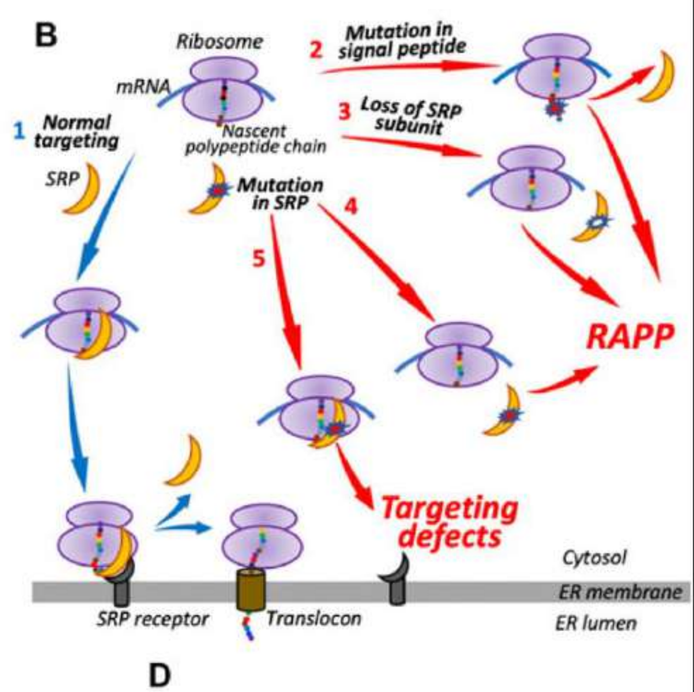
SRP targets proteins using a series of events called the SRP cycle. SRP binds N-terminal signal sequence of the nascent polypeptide chain of secretory protein (chain of colored circles), pauses the translation and targets ribosome-nascent chain complex to SRP receptor (purple) in ER membrane. Interactions between SRP and SRP receptor lead to transfer of paused ribosomes to the **translocon** (red cylinder) followed by release of SRP and continuation of protein synthesis. The nascent chain is translocated through the ER membrane into the lumen for further transport and modifications.

It is estimated that more than 30% of eukaryotic proteins are **secretory or membrane proteins**. The secretory (secreted out from the cell) and membrane proteins are transported within the vesicles. So they are stored in ER until **transported in vesicles**. Thus, defects in SRP biogenesis and mutations in SRP subunits may have pathological effects on a large number of these substrates leading to human diseases



Consequence of defect in SRP

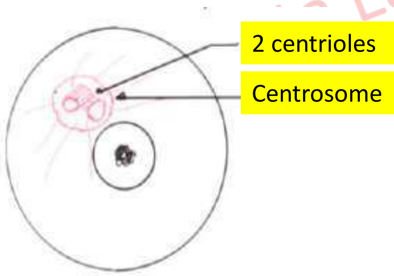
If SRP recognizes the substrates but is unable to target them to ER, they may mislocalize or degrade. All these events lead to dramatic consequence for protein biogenesis, activating protein quality control pathways, and creating pressure on cell physiology, and might lead to the pathogenesis of disease. Indeed, SRP dysfunction is involved in many different human diseases, including: congenital neutropenia; idiopathic inflammatory myopathy; viral, protozoal, and prion infection; and cancer. SRP failure can cause various diseases

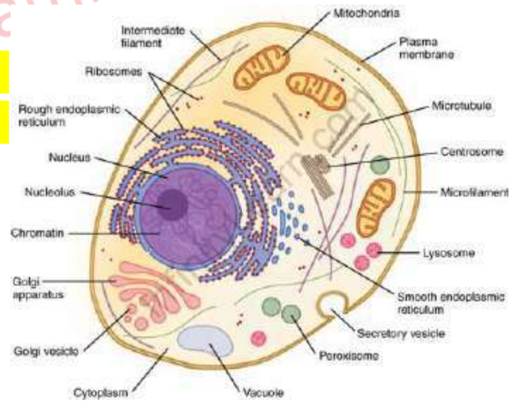


Centrosome

Centrioles are found in centrosomes. The centrosome is made up of **two perpendicular centrioles**, linked together by **interconnecting fiber, 100 of proteins and matrix**. Centrosome undergo duplication once every cell cycle so that their number remains stable, like the genetic material of a cell. The centrosome is positioned in the cytoplasm outside the nucleus but often near to it. Centrosome in the animal cells is very much like DNA. During cell division, **one centrosome** from the parent cell is transferred to each daughter cell. Centrosome consist of paired structures called centrioles, lying at ***right angles*** to one another. It is found only in animal cells and some lower plants (e.g. chlamydomonas).

Centrosomes/Centrioles are found in almost all eukaryotic animal cells, protozoan protists (except some forms like Amoeba), some fungi and the cells of all those eukaryotic plants where flagellate structures are present in the life cycle (many green algae, bryophytes, pteridophytes and cycads). They are absent in angiosperms, higher gymnosperms, some algae and fungi.





Assignment: Which structure contain flagella in algae, bryophytes, pteridophytes and cycads? (All within one week)

Structure of Centrioles

Centrioles are cylindrical, microtubular structures. With a diameter of about 250nm and a length ranging from 150-500nm in vertebrates, centrioles are some of the largest protein-based structures. They are visible under light microscope, but the details of centriole structure were revealed only under electron microscope. Usually two centrioles are found associated together in human and higher eukaryotes. The two centrioles are at right angles to each other. The pair of centrioles is often called **diplosome.**

Structure of a Centriole

A centriole is made of nine sets of microtubules. Each set consists of **three microtubules known as triplet microtubules.** The arrangement is known as 9+0, unlike cilia and flagella that contains doublet microtubules in a 9+2 arrangement. Triplet microtubules are very strong because they are composed of three concentric rings of microtubules that form together. **Triplet microtubules are seen in other strong microtubules structures, such as the basal bodies of cilia and flagella**. Each triplet is bonded together by special proteins that give a centriole its shape. Surrounding the triplet microtubules is an amorphous material called **pericentriolar material**, which contains many of the molecules necessary for the construction of microtubules. Each microtubule is made up of small units of **tubulin,** a small monomer that can join together to create long, hollow tubes that resemble straws.

A centriole possesses a whorl of **nine peripheral fibrils**. Fibrils are absent in the center. The arrangement is, therefore, called 9 + 0. Fibrils run parallel to one another but at an angle of 40°. Each fibril is made up of **three sub-fibers**. Therefore, it is called **triplet fibril**. The three sub-fibers are in reality microtubules joined together by their margins and, therefore, sharing the common walls made of 2-3 proto-filaments. From outside to inside the three sub-fibers of a triplet fibril are named as C, B and A. Sub-fiber A is complete with 13 proto-filaments while B and C sub-fibers are incomplete due to sharing of some microfilaments.

Structure of Centrioles

The center of centriole possesses a rod-shaped proteinaceous mass known as **hub**. From the hub, develops 9 proteinaceous strands towards the peripheral **triplet fibrils**. They are called **spokes**. Due to the presence of radial spokes and peripheral fibrils, the centriole gives a **cart wheel appearance** in T.S.

On the outside of centriole are present dense, amorphous, protoplasmic plaques in one or more series. They are called **massules or pericentriolar satellites**. Their position is changeable with the different states of the cell. Massules act as nucleating centres for the growth of microtubules during **aster formation** and **production of more centrioles** (during G2 phase).

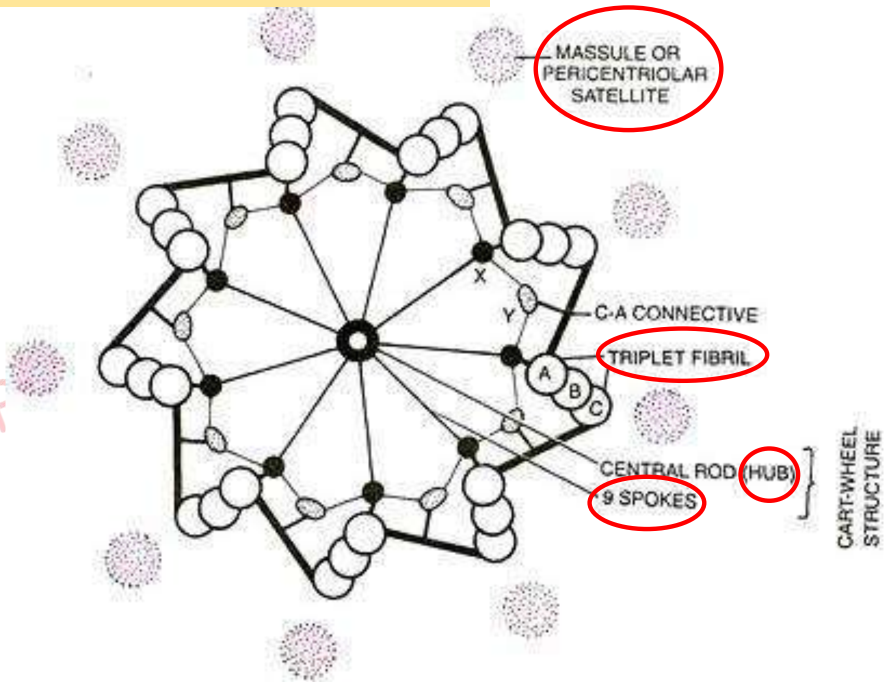
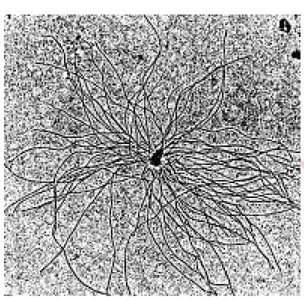


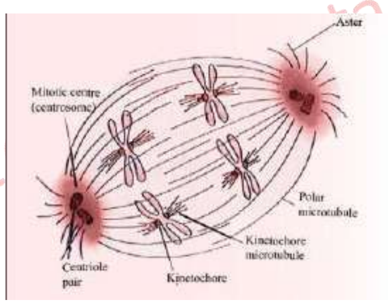
Fig: Ultrastructure of centriole as seen in T.S.

Role of centriole in cell division:

Separation of chromosome in cell division

Centrioles usually exist in a pair (mother and daughter) within the cell's centrosome, located near the nucleus, where they help organize the mitotic spindle during cell division. Centrioles are capable of **replication.** It is duplicated during S phase of the cell cycle and competes in G2 phase before M phase. Just before mitosis (M phase), the two centrosomes move apart until they are on opposite sides of the nucleus. As mitosis proceeds, microtubules grow out from each centrosome with their plus ends growing toward the metaphase plate. These clusters of microtubules are called spindle fibers. Centriole replication also occurs at the time of formation of basal bodies of cilia and flagella.

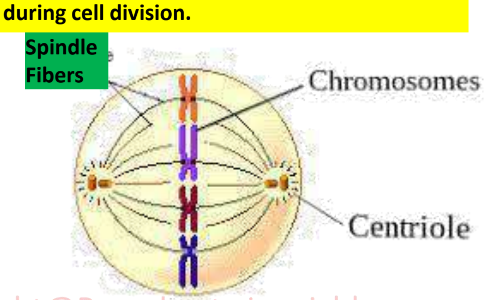




The figure shows the microtubules growing *in-vitro* from an isolated centrosome. The centrosome was supplied with a mixture of alpha and beta tubulin monomers spontaneously assembled into microtubules only in the presence of centrosomes.

Aster – **structure of microtubule that reradiates form the centrosome**

Spindle fiber – **microtubules that form a spindle-like structure and attach to chromosomes to pull them apart during cell division.**



Role of Centrioles in formation of cilia, flagella

Basal bodies

Cilia and **flagella** are projections from the cell. They are motile and designed either to **move the cell** itself or to **move substances over or around the cell**. Cilia and flagella have the same internal structure. The major difference is in their length. Specific type of cell that need flagella is sperm cell which help them to move. The centriole forms the basal body of the flagella of sperm from where microtubules are developed. Absence of centriole in sperm will lead to failure in formation of flagella without which the sperm cannot move.

They are made up of **microtubules** and covered by an extension of the plasma membrane. This microtubule-based structures of cilia is known as **axoneme**. From this '**basal body**' the growth and operation of the microtubules in a cilium or flagellum occurs.

Basal bodies are structures that are associated with the flagella and cilia in the eukaryotic cell. Eukaryotic cilia and flagella are built from basal bodies which are similar to centrioles. A basal body is a **single, modified centriole** that is found at the base of a cilium or flagellum of eukaryotic cells. It is not a pair of centrioles. The structure is made up of protein subunits.

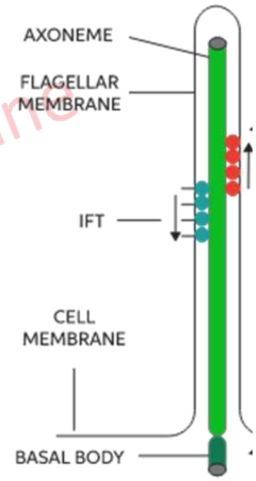
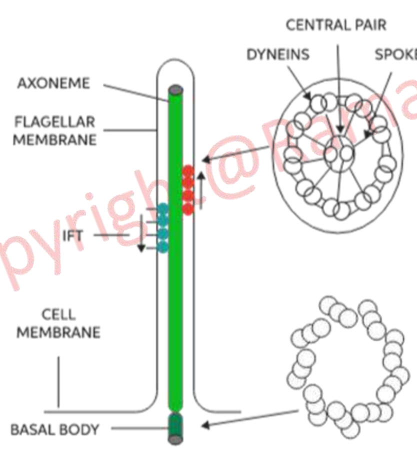


Fig: Flagella

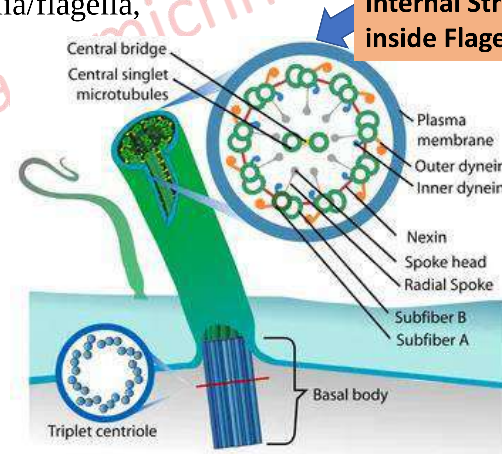
Role of Centrioles in formation of cilia, flagella

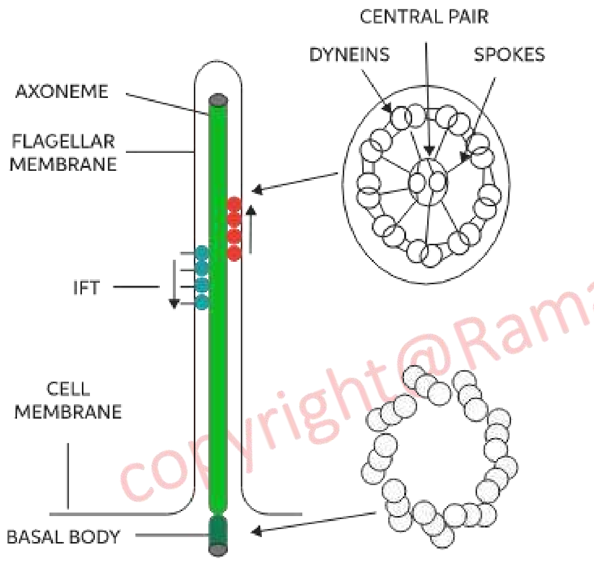
Basal bodies

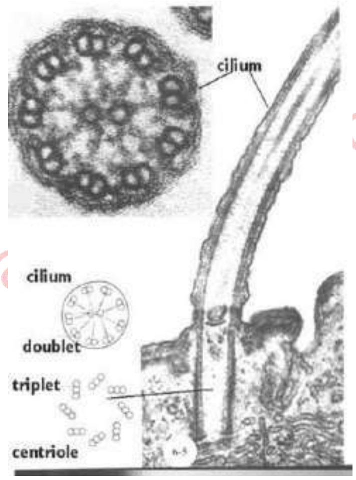
A basal body is structurally identical to a single centriole, typically composed of nine sets of triplet microtubules arranged in a cylinder (a 9x3 arrangement) and **doublet** in the center. The basal body has a **9+0** arrangement of nine triplet microtubules in a cylindrical, wheel-like structure, while the axoneme, which extends from the basal body, has a **9+2** arrangement of nine outer doublet microtubules surrounding a central pair of single microtubules in motile cilia/flagella, or a **9+0** arrangement in non-motile cilia.

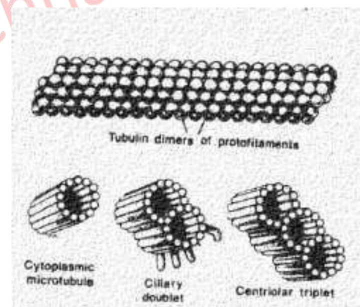


Internal Structure inside Flagella







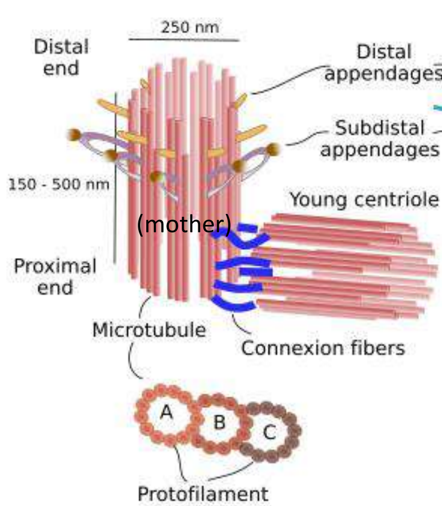


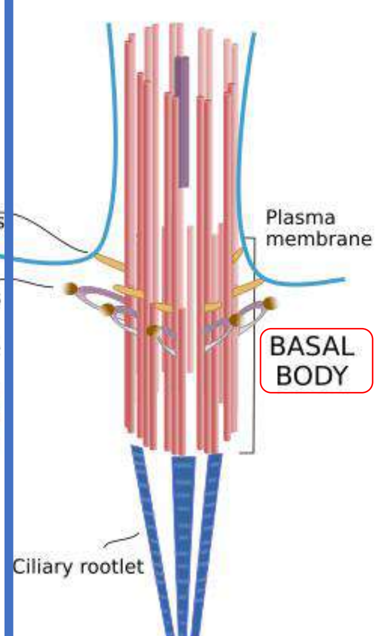
Mother and daughter centriole

CENTRIOLE

Mature (mother)

Young = daughter





BASAL BODY

Types of cilia

There are two types of cilia that include:

Primary cilia (non-motile cilia) -. The majority of cells, however, have primary cilia. Because they **lack** a central pair of microtubules (9+0), primary cilia are **incapable of motility** and are described as paralyzed in some books (meaning that they are not capable of motility). Some of these cilia do not protrude beyond the surface of the cell because they are very **short tubular Most**. Most of then have sensory function. Eg. cells of the kidney and photoreceptors on the external segments **of rods and cones** of the retina, olfactory receptors in **cilia of their dendrites**.

Secondary (Motile cilia) - Almost all motile (secondary) cilia and flagella have the same internal structure and have **essentially the same function**. Whereas flagella are generally few in number (< 5) and relatively long, cilia are typically short and are present in many copies (> 100) in a cell. The structure of both is based on a **"9 + 2"** arrangement of microtubules in which there are nine pairs of microtubules around the perimeter, with two single microtubules, which form the central pair. Attached to these microtubules are basal bodies. This process can either result in the cell moving through the water, typical for many single-celled organisms, or in moving water and its contents across the surface of the cell. In the human body, only a few cells have motile cilia. Some of these include sperm cells and ependymal cells located in brain vesicles. The function of some cilia is to move water relative to the cell in a regular movement of the cilia. It comprises of epithelial cells that line the human respiratory tract, where cilia constantly move mucus up from the lungs to the back of the throat. The cilia in the fallopian tubes, move the eggs down the tube toward the uterus. Scientists have known for some time that defects in ciliary motility lead to specific disorders in humans, such as :

Sterility in the instance of defective sperm cilia

Situs inversus in which defective embryonic cilia cause internal organs such as the heart to end up on the wrong

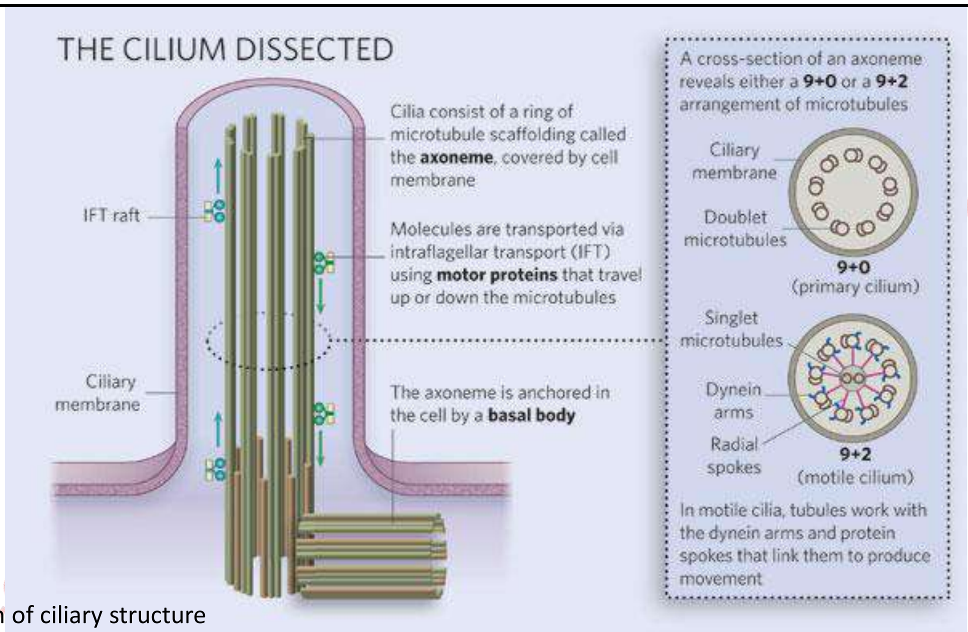


Figure 1: Diagram of ciliary structure

The organelle is membrane-bound and contains multiple microtubules running along its length. Whereas primary cilia have relatively little additional structure, motile cilia have both a central doublet of microtubules as well as inner and outer dynein arms and radial spokes, which are all needed for motility.

Role of Primary cilia

Cilia as Sensory Receptors

If primary cilia cannot move, what do they do? Defects in primary cilia can cause diseases, which suggests that they have an active role to play, but what is this role? Careful examination of kidney tubule cells showed that the primary cilia bend when exposed to moving liquid (as would be the case in working kidneys). Cells that were exposed to flowing liquid showed a very specific response as shown in the figure on the right. The results were interpreted to mean that the normal primary cilia were acting as sensors called mechanoreceptors, responding to flow by opening calcium channels and allowing a rapid influx of calcium ions into the cells.

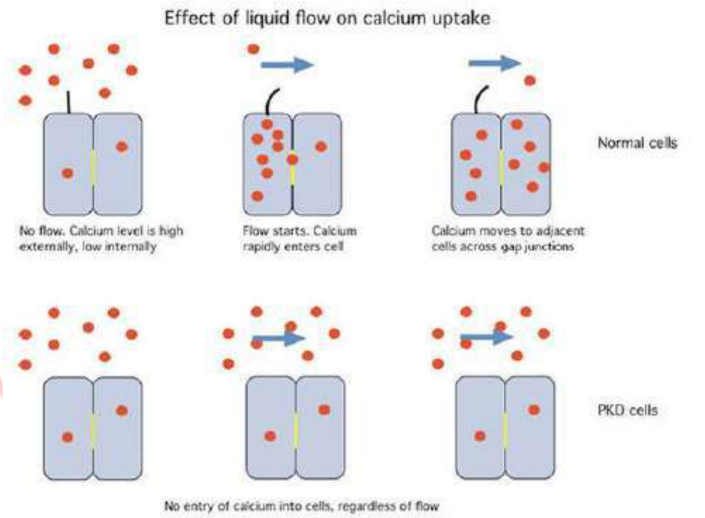


Figure : The effect of fluid flow on the uptake of calcium in normal and PKD (polycystic kidney disease) kidney cells. The primary cilia in normal cells respond to the mechanical signal produced by the flowing liquid to initiate rapid uptake of calcium. PKD cells, which lack primary cilia, show no response to flow. It results in deregulated calcium signaling, an impairment in the normal flow-induced influx of calcium into the renal tubular cells. It may lead to cyst formation in the kidney.

Ciliopathies – cilia related disorders

Findings on Centrioles:

As a result, the structure and function of the centriole, the centrosome, and the basal body have an impact upon many aspects of development and physiology, resulting in diseases that can readily be modeled in Drosophila. Such diseases include many forms of cancer, where centrosome aberrations have been known for over a century, and a variety of heritable diseases, where developmental disorders characterized by defects in cilia (ciliopathies) or asymmetric cell divisions in the brain (microcephaly) are associated with cilia, centriole, and centrosome defects.

In ciliated or flagellated cells, centrioles also function as basal bodies, structures anchored below the plasma membrane to seed axonemes, the MT-based structure of cilia and flagella. In recent years, evidence has accumulated for an indispensable role for cilia and flagella in various cellular and developmental processes, motility, propagation of morphogenetic signals in embryogenesis and sensory perception.

Q. Where do we find centrioles ?

Centrioles are present in (1) animal cells and (2) the basal region of cilia and flagella in animals and lower plants (e.g. chlamydomonas). In cilia and flagella centrioles are called 'basal bodies' but the two can be considered inter-convertible.

.Several examples demonstrate that basal bodies and centrioles are interchangeable. Centrioles are absent from the cells of higher plants. The primary purpose of cilia in mammalian cells is to move fluid, mucous, or cells over their surface.

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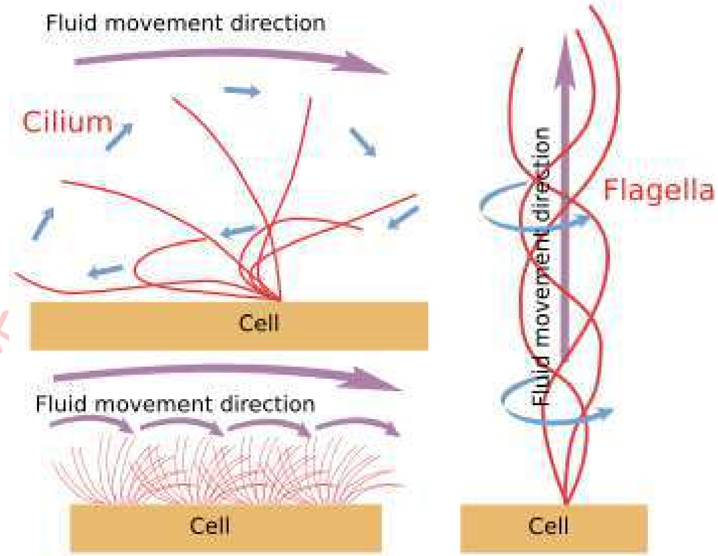
Assignment: Presentation on: Centrioles related diseases (Roll no. ….)

Cilia and Flagella motion:

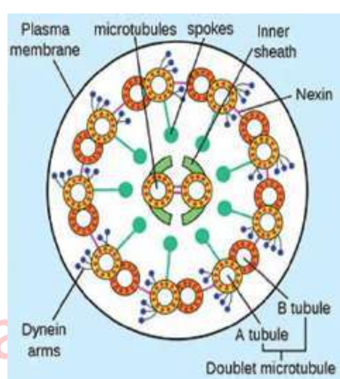
Flagella produce screw-like motions and thus a relative movement in the direction of the axis. In a bacterial cell, a flagellum twists in a circle like a screw. **The counter clockwise movement moves the organism forward** while the clockwise movement pulls it backwards. Eukaryotic flagella shows whip-like motion to move forward. In eukaryotes, flagella typically move in a **whip-like, bending, or undulating motion**. This motion is a back-and-forth movement, powered by ATP-driven protein motors (dynein arms) that cause the internal microtubules to slide past each other.

Cilia produces a whip-like moment thus causing a relatively lateral movement, for example in the respiratory cilia, and in the movement of the paramecium. Cilia movement is like beating, which impulses the liquid parallel to the cell surface.

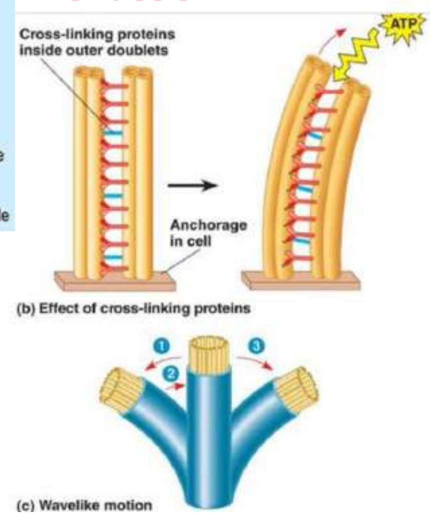


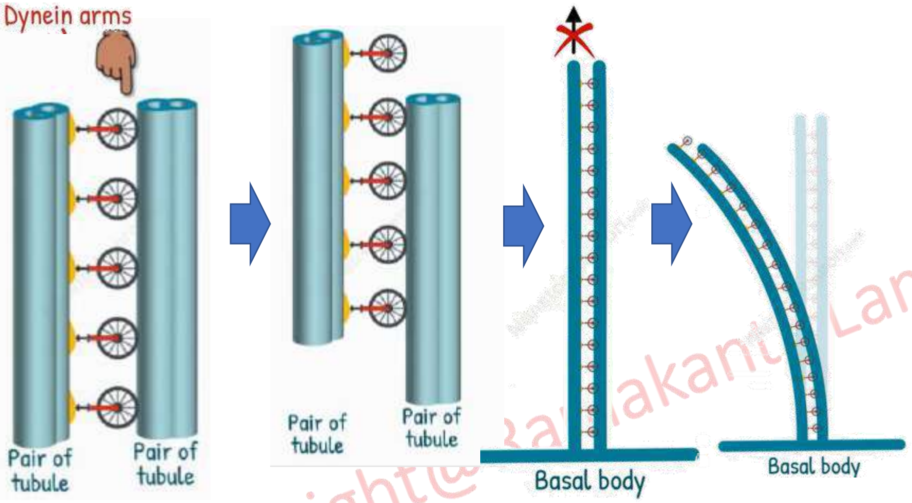
Physiology of ciliary movement

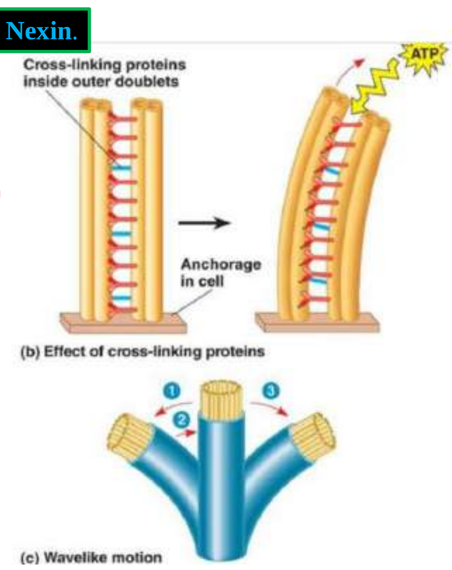
The axonemal elements of cilia provide the ciliary motor/motion, which is powered by ATP hydrolysis. Eukaryotic cilia and flagella have a common ultrastructure, **nine outer doublet** microtubules surrounding a pair of central microtubules, although their motility pattern is different from each other. There are water-propelling cilia such as mussel (a sea mollusk) gill cilia and mucus-propelling ones such as mammalian tracheal cilia. The main components of cilia and flagella are an **ATPase protein**, **dynein**, and the **constituent of microtubule, tubulin**.



The sliding is mediated by the **dynein arms**. In the presence of ATP, dynein functions with a set polarity to form transient (for short time) cross-bridges that cause the microtubule doublets of the axoneme to slide relative to one another. The microtubules are also connected by cross-linking proteins called **nexin**. So, when the microtubule tend to slide over one another, they are prevented by cross-linking proteins (nexin) which cause the microtubules to bend which results in formation of curvature in the microtubules. This ultimately results in bending of flagella or cilia and give movement.







Dynein protein is fixed at one side but can move or crawl at other side microtubule. As dynein protein crawls over another microtubule, 1st tries to move up but cannot do so as it is fixed to basal body and attached to 2nd microtubule by nexin protein (blue color)

Rather the sliding motion of dynein results in bending of the microtubule

Cilia and flagella: eukaryotes vs. prokaroytes

- Cilia ≠ pili! **Prokaryotes don't contain cilia**
  - Cilia: found on some eukaryotes, function = movement
  - Pili: found on some prokaryotes, function = attaching to surfaces
- Eukaryotic and prokaryotic flagella have different structure, motion